
Same Score, Different Strategy: Multi-Probe Diagnosis of Structural Fragility in LLM Reasoning

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Abstract

1 When two language models achieve the same score on a benchmark, are they
2 solving problems the same way? We show they are not: identical accuracy can
3 be reached through genuine computation or surface pattern recognition, and the
4 two strategies only become distinguishable *beyond* the accuracy number. We
5 apply three independent diagnostic probes—surface rename, cross-session plan-
6 vs-execution consistency, and training-data proximity—to 219 reasoning prob-
7 lems across arithmetic, planning, and algorithmic optimization, running five mod-
8 els (including the reasoning-trained o4-mini) for approximately 20,000 API
9 calls. Three-probe convergence labels apply most completely to the arithmetic
10 and algorithmic-optimization families; on Blocksworld our strict PDDL proto-
11 col aborts in 84–100% of sessions (itself a measurement-protocol finding), so the
12 Blocksworld execution probe is reported via a parsing-tolerant rerun with limited
13 convergence coverage. Three findings stand out. First, the same block-rename op-
14 eration *improves* two models, *destroys* one, and *leaves two untouched*—on iden-
15 tical problems. Second, weighted interval scheduling collapses every model to
16 zero under entity rename despite near-perfect canonical accuracy, isolating algo-
17 rithmic depth as a robustness axis orthogonal to benchmark score. Third, models
18 accept injected wrong intermediate states at high rates yet still recover the correct
19 final answer roughly half the time, showing that compliance and derailment are
20 empirically separable. None of these patterns is visible from benchmark accuracy
21 alone.

22 1 Introduction

23 Imagine two students score 80% on the same exam. You rename every person in every word prob-
24 lem; numbers and structure are unchanged. Student A still scores 80%; Student B drops to 0%.
25 Who actually understood the material? This is, roughly, what we observe in frontier LLMs: two
26 models can reach identical benchmark accuracy via fundamentally different strategies, and standard
27 accuracy reporting cannot tell them apart.

28 **The surprising finding.** On shortest-path planning, Claude Sonnet 4 scores 64.7% vs. GPT-4o’s
29 41.2%, then drops to 0.0% under pure entity renaming while GPT-4o retains 26.5% (matched SP
30 subset, $n=34$, Fisher $p=0.0021$). This is not a two-model anecdote: across Claude, GPT-4o, Gemini,
31 and Llama [Dubey et al., 2024], robustness ordering is family-dependent and cannot be inferred from
32 canonical accuracy. (A same-direction coin-change reversal at $n=10$ is reported as exploratory in
33 Appendix A.)

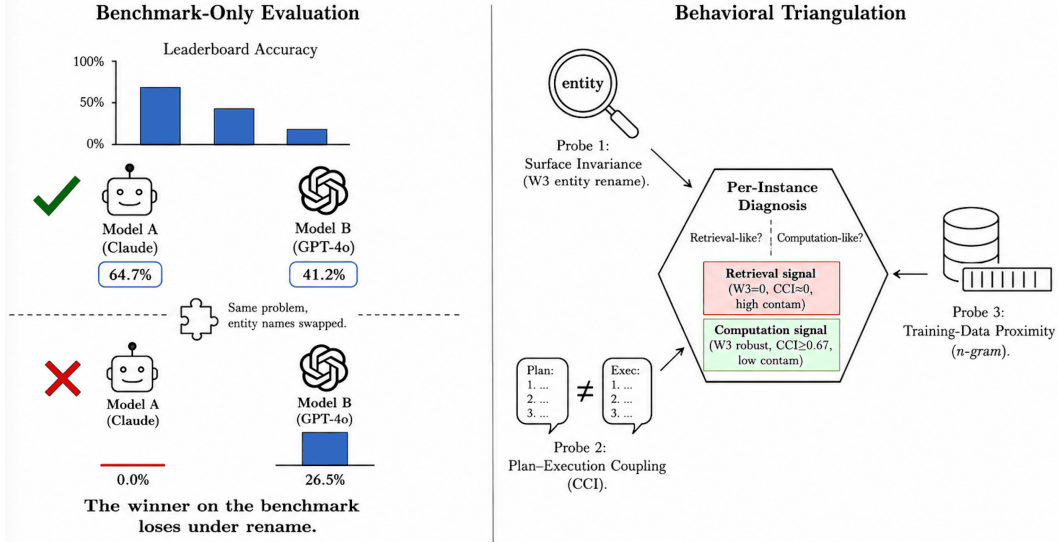


Figure 1: Multi-probe diagnostic evaluation framework. Three orthogonal probes apply to every problem instance; a per-instance label is issued only when probe signals converge (Section 3.1; labels are calibration points, not primary results). Full pipeline: Figure 21 (Appendix G).

34 **Why one test is not enough.** A drop under renaming alone admits mundane explanations (tok-
 35 enization, residual difficulty, noise). Our approach is *convergence*: three independent tests with
 36 non-overlapping assumptions. Prior work [Mirzadeh et al., 2025, Wu et al., 2024, Wang and Zhao,
 37 2024] reports group-level drops; we diagnose individual instances by asking whether all three sig-
 38 nals agree.

39 **What is genuinely new.** Two design choices (Table 1): cross-session plan-vs-execution under
 40 hard isolation (CCI; neither within-context [Turpin et al., 2023] nor own-CoT perturbation [Lanham
 41 et al., 2023]); and a built-in DP exposure control (CC/SP/WIS share algorithm, differ in public-
 42 corpus prevalence).

43 **The three probes.** Three orthogonal diagnostics—surface rename (W_3), cross-session plan-vs-
 44 execution (CCI/TEP under hard session isolation), and Infini-gram public-corpus proximity [Liu
 45 et al., 2024]—apply to every problem (Figure 1; metrics in Section 3.1; full pipeline in Ap-
 46 pendix G).¹

47 2 Related Work

48 **Surface sensitivity.** Mirzadeh et al. [2025] (GSM-Symbolic) show arithmetic accuracy drops un-
 49 der template-variable perturbation in a single domain at the group-mean level. Wu et al. [2024]
 50 extend this across 11 counterfactual tasks (again group-level). Wang and Zhao [2024] (RUPBench)
 51 apply nine perturbation types over 15 reasoning datasets. Li et al. [2024] (GSM-Plus) generate di-
 52 verse variants of GSM8K and find consistent accuracy drops; Huang et al. [2025] (MATH-Perturb)
 53 construct hard perturbations on competition math. We differ by coupling surface, execution, and
 54 exposure probes *per instance* on shared problems (Table 1), with a built-in DP exposure gradient
 55 (CC/SP/WIS) and matched-canonical pairs that invert across models.

56 **Most related concurrent work.** ? (MATCHA) test the decoupling hypothesis on GSM8K with
 57 *within-trace* answer-conditioned perturbations and an LLM judge—complementary to our determin-
 58 istic W_3 nonce-token rename and cross-session CCI. The Robust Reasoning Benchmark [Golikov
 59 et al., 2026] applies 13 structural perturbations to competition-math items at group scale but includes

¹Anonymous code and logs: <https://anonymous.4open.science/r/retrieval-vs-computation-EA8F/README.md>.

Table 1: Full-probe novelty map. W1 paraphrase, W2 reformat, W3 entity rename, W4 formal notation, W5 direction reversal, W6 new numbers; P2A cross-session plan-vs-execution, P2B mid-solve false-state injection; P3-I public-corpus n -gram proximity (Infini-gram), P3-M mechanistic final-rank analysis; Conv. per-instance convergence label combining all probes. MATCHA, RRB, and PutnamGAP each overlap one or two columns; this work uniquely combines all columns on the same instances with a built-in training-exposure control.

Related work	Probe 1: Surface variants						Probe 2: Exec.		Probe 3: Exp.		
	W1	W2	W3	W4	W5	W6	P2A	P2B	P3-I	P3-M	Conv.
GSM-Symbolic [Mirzadeh et al., 2025]						✓					
GSM-Plus [Li et al., 2024]	✓	✓				✓					
MATH-Perturb [Huang et al., 2025]						✓					
RUPBench [Wang and Zhao, 2024]	✓	✓	✓								
Wu et al. [Wu et al., 2024]						✓					
Turpin et al. [Turpin et al., 2023]							✓				
Lanham et al. [Lanham et al., 2023]	✓						✓	✓			
Saparov & He [Saparov and He, 2023]							✓				
Lightman et al. [Lightman et al., 2024]							✓				
Sharma et al. [Sharma et al., 2024]								✓			
Razeghi et al. [Razeghi et al., 2022]										✓	
Golchin & Surdeanu [Golchin and Surdeanu, 2024]										✓	
Shi et al. [Shi et al., 2024]										✓	
MATCHA [Jiang et al., 2025]							✓				
RRB [Golikov et al., 2026]	✓	✓	✓			✓					
PutnamGAP [Hao et al., 2025]			✓			✓					
This work	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

no execution or contamination probes. PutnamGAP [Hao et al., 2025] stress-tests surface renames and kernel rewrites on thousands of Putnam problems without per-instance execution or proximity controls. We position against all three in Table 1: our contribution is per-instance triangulation on a fixed 219-problem bank with session-isolated Probe 2 and Infini-gram proximity on the same instances.

CoT faithfulness. Prior work [Turpin et al., 2023, Lanham et al., 2023, Saparov and He, 2023, Lightman et al., 2024, Tutek et al., 2025] intervenes within a single trace or on model weights. Our CCI compares a session-A declaration to session-B execution with no shared context ($\epsilon=0.01$ numeric match)—behavioural plan-execution consistency under hard isolation, not within-trace or parametric faithfulness.

Training-data contamination. Razeghi et al. [2022] show pretraining term frequency predicts arithmetic accuracy. Golchin and Surdeanu [2024] detect contamination via guided instruction completion. Shi et al. [2024] extend contamination analysis to reasoning benchmarks. For closed-weight models we treat Infini-gram overlap [Liu et al., 2024] as a public-corpus proximity proxy; we discuss its limitations and partial validation via the open-weight Llama model in Section 4.3.

LLM planning and execution. Valmeekam et al. [2023a] demonstrate systematic Blocksworld planning failure. We extend PlanBench [Valmeekam et al., 2023b] with a cross-session execution probe and a false-state injection probe, testing whether models that accept an injected error maintain correct final-answer generation.

Concurrent work on reasoning-model robustness. von Recum et al. [2026] perturb models’ own CoT tokens within-trace; our Probe-2B injects external wrong state into fresh sessions with no prior reasoning context—complementary tests of compliance vs. final correctness.

3 Setup

We test five models—four frontier chat models and the reasoning-trained o4-mini—on three types of reasoning problems, each presented in up to seven surface forms. Every problem has a veri-

85 fied correct answer, and every surface variant preserves that answer exactly; only the presentation
 86 changes. The core question: does the model still get it right?

87 **Problem families.** Three reasoning domains, chosen to dissociate three confounds that would
 88 otherwise undermine the probes individually: (i) **GSM** (arithmetic) calibrates the probes against a
 89 benchmark with documented contamination structure but a very different algorithmic style; (ii) **BW**
 90 (planning) exposes the format-vs-policy confound (“did the model actually fail, or did the parser?”)
 91 and provides the matched standard/mystery contrast that directly probes lexical content while hold-
 92 ing planning structure fixed; (iii) **ALGO** (algorithmic optimisation) is the only family where we can
 93 hold the underlying algorithm fixed (dynamic programming, DP) while varying training-text expo-
 94 sure across three orders of magnitude—this is what makes the same-algorithm vs. different-exposure
 95 control in Section 4.3 possible at all.

96 GSM (44 GSM8K problems [Cobbe et al., 2021]): arithmetic calibration with documented contam-
 97 ination structure. BW (65 PlanBench [Valmeekam et al., 2023b] instances): planning with stan-
 98 dard/mystery lexical contrasts. ALGO (110 DP problems): three subtypes with shared algorithm
 99 but divergent public-corpus exposure—

- 100 • **Coin change (CC):** minimum number of coins summing to a target. Very common in
 101 programming tutorials online.
- 102 • **Shortest path (SP):** minimum-cost route through a weighted graph. Moderately common.
- 103 • **Weighted interval scheduling (WIS):** maximum-value set of non-overlapping intervals.
 104 Rarely phrased this way in public text.

105 Per-instance audit logs are released (Appendix H). GSM calibrates all probes; ALGO holds DP
 106 fixed while varying exposure; BW stress-tests interactive measurement (Section 4.4). Coverage:
 107 Table 2. Throughout, BW Probe-1 *accuracy* is scoped to the $n=65$ bank problems; within-model
 108 per-problem correlations (Figure 11) instead pool canonical and variant instances per model, which
 109 is why their BW n is larger (up to 109).

110 **Variants and models.** Each canonical problem has six verified variants (W1–W6; Appendix B);
 111 W3 renames all entities to nonce tokens. Five models at $T=0$ zero-shot CoT via OpenRouter: Claude
 112 Sonnet 4 (hereafter Claude), GPT-4o, Gemini 2.5 Flash (Gemini), Llama-3.1-8B-Instruct (Llama,
 113 Dubey et al., 2024), and o4-mini. We use these short names throughout. o4-mini was queried
 114 in its default API mode (no explicit reasoning-token budget set; chain-of-thought is internal and
 115 not exposed in the response). o4-mini reaches 1.00 ALGO / 0.841 GSM canonical yet is most
 116 rename-fragile on BW ($W_3=0.185$)—family-specific weakness invisible from accuracy alone.

117 3.1 Metrics and convergence labels

118 Each probe excludes confounds the others admit; labels require convergence. $R_{W_3} = \text{Acc}_{W_3} / \text{Acc}_{\text{can}}$;
 119 VRI compares W3 to vocabulary-preserving variants. CCI (Cross-session Consistency Index) is the
 120 fraction of declared numeric steps matching execution ($\varepsilon=0.01$); TEP (Trajectory Error Propagation)
 121 and injection compliance measure Probe-2B derailment. Infini-gram overlap on The Pile/DCLM is
 122 a public-corpus proximity proxy [Liu et al., 2024]. Instances are *retrieval-consistent* only when all
 123 three signals align (and *computation-consistent* when all reverse); under strict rules $\sim 62\%$ of ALGO
 124 instances stay ambiguous (Section 4.5). The 12 high-confidence labels are calibration points for the
 125 probe design—showing the signals can co-align—not primary results; Sections 4.1–4.4 report the
 126 behavioural findings.

127 **CCI interpretability.** Phase 1 (full-plan declaration) and Phase 2A (per-step execution) use dif-
 128 ferent prompt formats by design; CCI therefore confounds inconsistent reasoning with differential
 129 format propensity and parser coverage. Among GPT-4o’s zero-CCI sessions, 17/29 lack extractable
 130 declared numeric steps and therefore receive CCI=0 *by construction*, not by reasoning failure. Du-
 131 plicate Phase-2A calls at $T=0$ on an audit subset yield identical outputs for Claude, ruling out API
 132 non-determinism as the driver of CCI spread. We report CCI as a descriptive fingerprint; a prompt-
 133 matched ablation is deferred (Limitations).

Table 2: Probe coverage. Ph1=declaration; Ph2A=execution; Ph2B= injection. ^aBW Phase 1 plans declared; CCI often non-computable due to illegal-action abort loops. ^b84% of BW Phase-2A sessions abort under the strict parser (Section 4.4); the NL-tolerant rerun yields non-zero solves on standard BW. In total the study comprises approximately 20,000 model API calls, spanning Probe-1 variant evaluation, Probe-2 (Phase 1/Phase-2A/Phase-2B), step-level reasoning-type classification, and variant generation/verification; Infini-gram index queries (Probe 3) are computed once per problem and are not counted in this total.

Family	P1 Vars	P2-Ph1	P2-Ph2A	P2-Ph2B	P3
GSM (5 mod.)	✓	✓	✓	✓	✓
ALGO (5 mod.)	✓	✓	✓ ^b	✓	✓
BW (5 mod.)	✓	✓ ^a	^b	—	✓
Mech.	Qwen-2.5-7B pilot (Appendix F)				

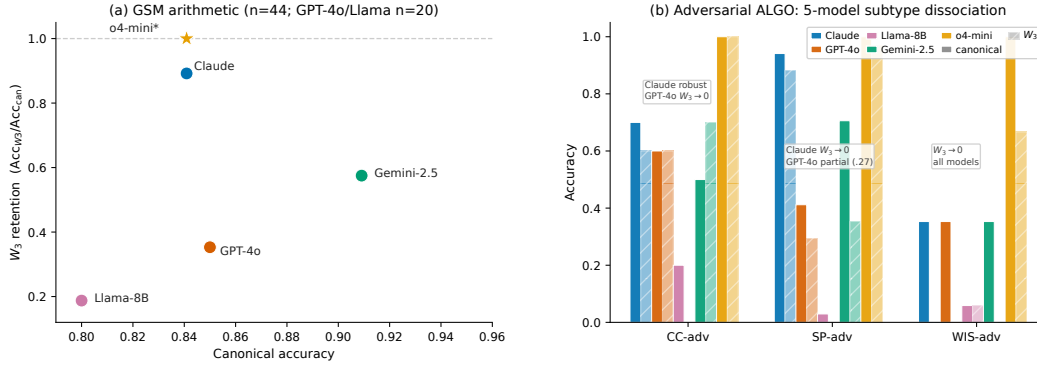


Figure 2: Accuracy-robustness dissociation across five models. (a) GSM calibration: Claude/Gemini/o4-mini at $n=44$; GPT-4o/Llama at $n=20$ bank-valid (partial replication; wide CIs). (b) Adversarial ALGO: SP-adv ($n=34$, Fisher $p=0.0021$) collapses Claude ($W_3=0$) while GPT-4o retains 26.5%; CC-adv reverses the pair; WIS-adv floors every model.

4 Results

4.1 Subtype-specific robustness inversion: the higher-accuracy model is not always the more robust one

Figure 2 and Table 3: on GSM, Claude/Gemini/o4-mini ($n=44$) span canonical accuracy 0.841–0.909 with divergent W_3 retention ($R_{W_3} = .892/.575/1.00$). GPT-4o and Llama replicate the pattern directionally at $n=20$ only (Wilson CIs span ~ 30 pp; Appendix A)—we treat them as partial replication, not powered five-model evidence. The adequately powered headline is ALGO SP-adv below.

The pattern is sharper on algorithmic problems. Shortest-path: Claude 64.7% canonical ([0.48,0.79]) collapses to 0.0% under renaming ([0.00,0.10]); GPT-4o 41.2% ([0.26,0.58]) retains 26.5% ([0.15,0.43]). The *better* model on the canonical is the *worse* one under rename. Coin-change reverses the direction: Claude 60% retention ([0.31,0.83]), GPT-4o 0% ([0.00,0.28]). Robustness ordering depends on the problem subtype.

Statistical confidence. SP matched-canonical ($n=34$): Claude 0/34 vs. GPT-4o 9/34 (Fisher $p=0.0021$); CC ($n=10$): Claude 6/10 vs. GPT-4o 0/10 ($p=0.0108$; exploratory). Within-model phi: ALGO $\phi_{GPT-4o}=+0.43$ (Fisher $p<0.001$); GSM $\phi_{o4-mini}=+0.66$ ($p<0.001$; Appendix D, Figure 11). Population-level Spearman is n.s. (Appendix D); ALGO model-rank retention trends positive ($\rho=+0.90$, $p=0.04$, $n=5$; Holm n.s.)—the dissociation is per-(model, subtype), not a universal axis.

Table 3: GSM Probe 1 ($n=44$ for Claude/Gemini/o4-mini; $n=20$ partial for GPT-4o/Llama; 95% Wilson CIs). The headline inversion is powered on ALGO SP-adv ($n=34$; Section 4.1); GSM rows calibrate near-matched canonical accuracy (0.800–0.850) with wide CIs for the $n=20$ cells.

Model	Acc_{can}	Acc_{W_3}	Δ	R_{W_3}
Claude	.841 [.71,.92]	.750 [.61,.85]	.091	.892
GPT-4o	.850 [.64,.95]	.300 [.15,.52]	.550	.353
Llama-8B	.800 [.58,.92]	.150 [.05,.36]	.650	.188
Gemini-2.5	.909 [.79,.96]	.523 [.38,.66]	.386	.575
o4-mini	.841 [.71,.92]	.841 [.71,.92]	.000	1.000

Table 4: GSM Probe 2 ($n=44$ /model; 43/44 parseable for o4-mini). Acc_{P2A} cells below are the phase2a-or-phase1 disjunction, **not** Phase-2A accuracy, and are **withdrawn** pending a re-run that persists phase2a_values (see results/README.md). Numbers are retained for provenance. CCI: deterministic step-value agreement ($\epsilon=0.01$). o4-mini is the strongest executor (.955) yet ranks mid-pack on CCI—strong execution without strong plan-following. Wilcoxon Claude vs. GPT-4o: $W=396$, $p=0.0068$.

Model	Acc_{P2A}	CCI mean	CCI med.	TEP
Claude	.864	.231	.216	.539
GPT-4o	.705	.108	.000	.599
Llama-8B	.455	.167	.000	.773
Gemini-2.5	.886	.270	.250	.652
o4-mini	.955	.220	.143	.628

153 **Variant asymmetry and WIS.** W6 (new numbers) stays near canonical while W_3 (rename) drops
154 for higher-retention models (Figure 10); 26/110 ALGO problems are universally W_3 -fragile (12 SP,
155 11 WIS, 3 CC). The $3 \times 7 \times 5$ grid (Figure 16): every model reaching $\geq 95\%$ WIS canonical collapses
156 to 0% under W_3 (including o4-mini 1.00 $\rightarrow$ 0.00); SP W5 (source/destination reversal) floors the
157 suite ($\leq 4\%$). Algorithmic depth is orthogonal to accuracy.

158 4.2 Models that declare a plan often do something completely different—and still get the 159 right answer

160 CCI is a behavioral fingerprint, not a performance predictor. Claude matches $\sim 23\%$ of declared
161 numeric steps vs. GPT-4o’s $\sim 11\%$ (median=0; Wilcoxon $W=396$, $p=0.0068$, $n=44$ zero-imputed
162 pairs), yet GPT-4o still answers correctly 70.5% of the time. Declaration-following does not predict
163 Claude’s correctness ($\rho=0.14$, $p=0.38$). Cross-model zero-CCI rates: Gemini 10/44 (23%), Claude
164 14/44 (32%), Llama 24/44 (55%), GPT-4o 29/44 (66%); among GPT-4o’s 29 zero-CCI sessions,
165 17/29 declared no extractable steps and 12/29 declared then diverged—final accuracy is nearly iden-
166 tical in both (0.706 vs. 0.667, n.s.), confirming step declaration and answer production are largely
167 independent. Claude and GPT-4o are different *kinds* of reasoners; CCI is a fingerprint, not a score.
168 TEP (Trajectory Error Propagation, Table 4) is high across all models (mean 0.54–0.77): the in-
169 jected wrong value propagates and alters many subsequent steps, yet the final answer still survives
170 at near-baseline rates—which is exactly the compliance $\neq$ correctness finding reported below.

171 **Injection compliance $\neq$ final correctness.** On 61 ALGO injection instances per model, accep-
172 tance is high (Claude 88.5%, GPT-4o 93.4%, Llama 39.3%, o4-mini 100%; Gemini 0% compliant)
173 yet post-injection accuracy stays near uninterrupted rates (Claude 52.5% vs. 54.1%; GPT-4o 50.8%
174 vs. 55.7%; o4-mini 37.7% vs. 40.9%)—compliance and correctness are empirically separable (Fig-
175 ure 3c). The figure shows the full three-way split (compliant / partial-refusal / format-ignored):
176 Gemini format-ignores 100% of injections (treats the injected state as malformed input and proceeds
177 without it), accounting for its 0% compliant rate; Llama splits across compliant and partial-refusal.

178 **Two five-model nulls.** Plausible vs. implausible injection ($n=61$ paired): no model differs sig-
179 nificantly (Claude $\Delta=0.0pp$, $p=1.00$; GPT-4o +4.9pp, $p=0.50$; Llama +3.3pp, $p=0.69$; Gemini
180 –3.3pp, $p=0.64$; o4-mini +4.9pp, $p=0.48$; Figure 17). Algorithm elicitation raises invocation 10–

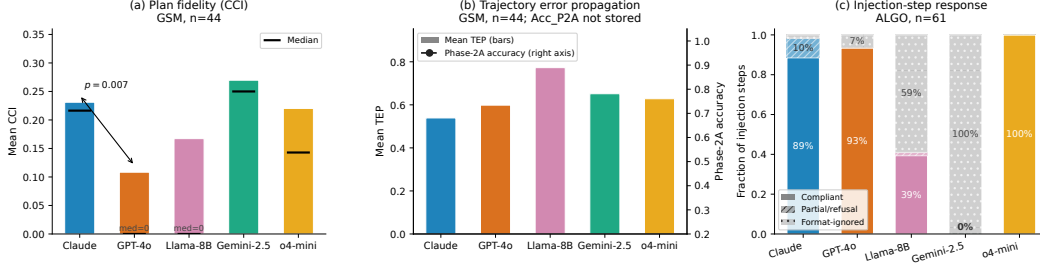


Figure 3: Probe 2 results. (a) Plan fidelity (CCI, GSM $n=44$): Claude has significantly higher numeric step agreement than GPT-4o (Wilcoxon $W=396$, $p=0.0068$); Gemini has the highest CCI (0.270); GPT-4o and Llama have median CCI=0; o4-mini sits between Llama and Claude (mean CCI 0.22, 43/44 parseable). Strong execution does not require strong plan fidelity. (b) Trajectory error propagation with Phase-2A accuracy overlay: TEP and accuracy are weakly coupled across the five-model pool. (c) Injection-step response (ALGO Phase 2B, $n=61$ paired problem instances per model): Claude and GPT-4o are largely compliant; Llama partial; Gemini format-ignores every injection (0% compliant); o4-mini complies on 100% of injections.

Table 5: Within-ALGO proximity gradient (challenging instances). Template proximity $\bar{c}_T$ from Infini-gram on The Pile/DCLM. WIS and CC share DP-solvable structure but differ in proximity; importantly, WIS canonical accuracy (~ 0.35) is also lower than CC (~ 0.65), so task hardness covaries with proximity—this is a partial control, not a clean isolation.

Subtype	$\bar{c}_T$	Claude		GPT-4o	
		Can	W3	Can	W3
CC (chall.)	0.468	.700	.600	.600	.000
SP (chall.)	0.147	.647	.000	.412	.265
WIS (chall.)	0.000	.353	.000	.353	.000

181 50 $\times$ but not accuracy in any model (mean $\Delta=-2.0$ pp; Fisher $p=0.21-0.83$): Claude 0.500 $\rightarrow$ 0.459,
 182 GPT-4o 0.500 $\rightarrow$ 0.518, Gemini 0.300 $\rightarrow$ 0.311, Llama 0.218 $\rightarrow$ 0.148, o4-mini 0.436 $\rightarrow$ 0.418.

183 4.3 Problems the model has likely seen before are easier—but fragile under name changes

184 At zero template proximity (WIS), W_3 robustness is zero for every model—consistent with both
 185 lower exposure and higher baseline difficulty; we cannot disentangle these from the current design
 186 (Table 5). CC retains partial robustness for Claude despite shared DP structure. Proximity-VRI
 187 correlations use the frozen 61-problem adversarial pool (34 SP + 10 CC + 17 WIS challenging IDs;
 188 earlier drafts said $n=64$). We report *instance-level* Pearson r vs. VRI (template-level r is signed
 189 differently for some models and is reported separately in the reproducibility bundle). Proximity
 190 predicts per-problem VRI for Claude ($r=+0.444$, $p=0.0003$) and GPT-4o ($r=+0.364$, $p=0.004$)
 191 but not Llama/Gemini ($r\approx 0.11-0.12$, n.s.). After residualizing on per-problem canonical accuracy,
 192 partial correlations survive for Claude ($r=+0.418$) and GPT-4o ($r=+0.380$; $n=61$), supporting a
 193 proximity-linked rename-fragility association within difficulty tier rather than a pure floor effect.
 194 Proximity also predicts Claude’s CCI rigidity ($r=+0.31$, $p=0.044$, $n=42$; 2 sessions dropped for
 195 missing proximity scores) independent of correctness ($r=-0.11$, n.s.). o4-mini breaks the link
 196 (instance $r=-0.171$, $n=61$; one-model observation). Figure 12 summarises macro metrics and
 197 correlations.

198 4.4 Blockworld: when the measurement protocol itself breaks down

199 Under interactive PDDL, 84–100% of BW Phase-2A sessions abort (Claude 16% / GPT-4o 0% /
 200 Llama 2% complete)—a protocol limit, not a model CCI result (Figure 18: Claude semantic va-
 201 lidity 0.68; GPT-4o format failures 25.6/session; Llama validity 0.27). The NL-tolerant rerun lifts

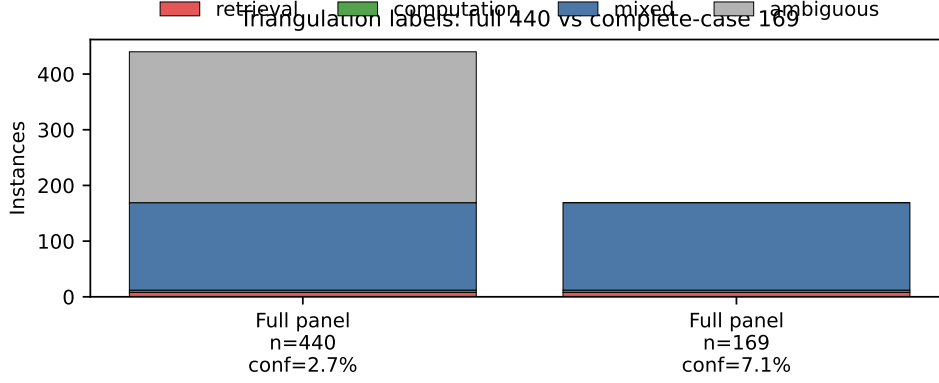


Figure 4: Executed-rule triangulation on the full 440 panel vs. the complete-case 169 subset (W3 + CCI + proximity present, parse OK). Missing Phase-2 / parse flags account for the ambiguous mass on the full panel.

standard-BW from 0/150 to 14/150 sessions (Fisher $p=8.9 \times 10^{-5}$);² Mystery-BW stays 0/45, aligning with the Qwen-2.5-7B mechanistic pilot [Qwen Team, 2024] (Appendix F). Paired Wilcoxon on W5-rename under the NL-tolerant protocol (Figure 19; note these differ from the strict-protocol values in Table 7): Claude 0.422→0.661 (+23.9pp, $p=1.0 \times 10^{-4}$); Gemini 0.385→0.569 (+18.5pp, $p=0.014$); Llama 0.321→0.101 (−22.0pp, $p<10^{-4}$); GPT-4o/o4-mini unchanged ($\Delta=0$ pp, n.s.).

4.5 When all three tests agree

Triangulation labels are calibration points, not primary results: they validate that the three probes *can* co-align on the same instance. The published headline uses the *executed* asymmetric five-field AND rule (named constants in the released triangulation code), *not* the appendix three-signal print. Under that executed rule on 440 ALGO instances we obtain 8 retrieval-consistent, 4 computation-consistent, 157 mixed, and 271 (61.6%) ambiguous; GSM (Claude, $n=44$): 35 computation-leaning, 9 ambiguous. Restricting to the complete-case subset where W3, CCI, and proximity are all present and the declaration parse succeeded ($n=169$) removes the missing-data pile: ambiguous drops from 271 to 0 while the 8/4/157 confident cells are unchanged (Figure 4; confident-label rate 2.7%→7.1%). Of the 157 mixed labels, 128 sit in a structural hole (W3=0 with contamination rank ≤ 0.5) where neither conjunction can fire (Figure 5). An appendix-style symmetric three-signal rule is a sensitivity analysis only: on the same 440 panel it yields 8/0/299/133 and flips 195/440 labels relative to the executed rule (Figure 7; Appendix J). Label rates remain threshold-sensitive under a 270-config sweep that collapses to 162 distinct configurations once binary W3 cutoffs are identified (Table 9). The primary findings are the behavioural patterns in Sections 4.1–4.4.

4.6 Intrusion errors under entity rename

Beyond accuracy drops, we score *intrusion* errors on W3 failures: answers that reuse an entity or numeric fragment from the canonical surface form after the rename (DS-02). GSM W3 error intrusion is 0/70 across models. On ALGO, rates are low for non-reasoning models (Claude 2.0%, GPT-4o 2.2%, Llama 0%, Gemini 4.9%) but elevated for o4-mini (5/43 = 11.6%, Wilson 95% CI [0.051, 0.245]; Figure 6). Fisher exact tests against other models: vs. Claude OR=6.45, $p=0.026$; vs. GPT-4o OR=5.86, $p=0.035$; vs. Llama OR= ∞ , $p=0.002$; vs. Gemini OR=2.57, $p=0.273$. We treat this as a secondary behavioural signature of reasoning-trained models under surface rename, not a primary claim.

²Standard-BW comprises 50 problems $\times$ 3 models run under the strict protocol = 150 sessions; Mystery-BW comprises 15 problems $\times$ 3 models = 45 sessions. Counts in this section are over *sessions*, not unique problems.

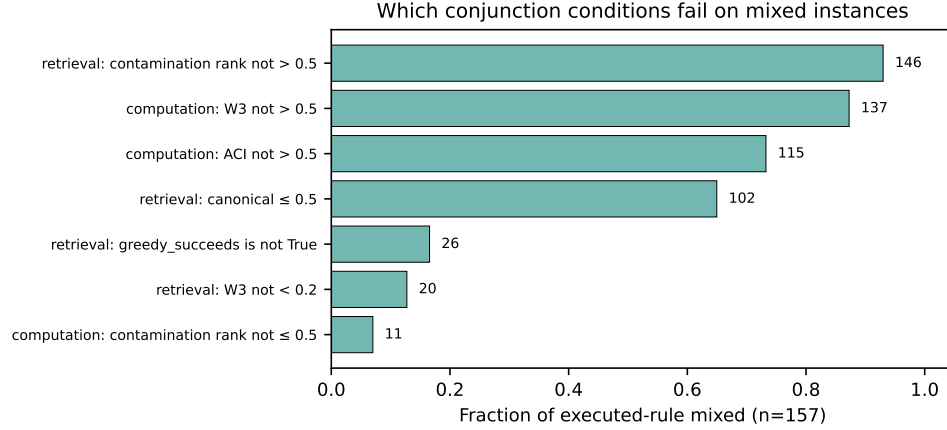


Figure 5: Which executed-rule conjunction conditions fail on the 157 mixed instances. The dominant pattern is the structural hole (low contamination rank with W3 fail), not W3-cutoff asymmetry.

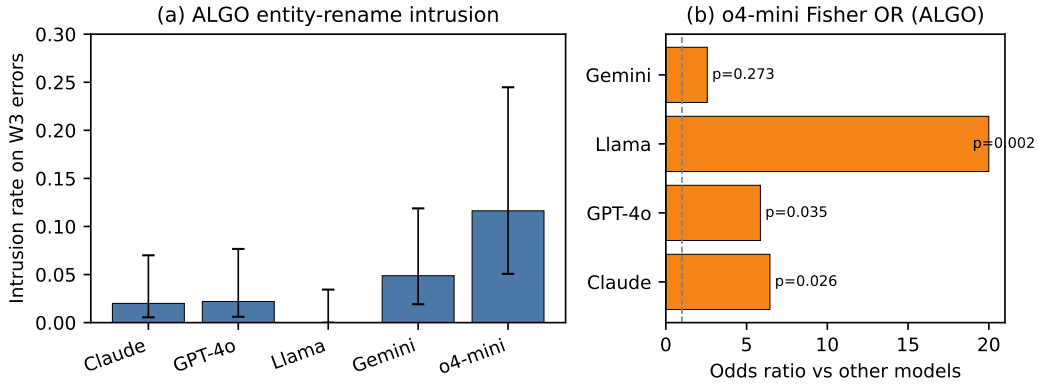


Figure 6: ALGO intrusion rates on W3 errors (Wilson 95% CIs) and Fisher odds ratios for o4-mini vs. each other model.

5 Conclusion

The five models form qualitatively different reasoners—Claude declares plans reliably (GSM CCI=0.231) without correctness coupling; GPT-4o rarely follows declarations (median CCI=0) yet recovers; Gemini peaks on canonicals (91%) but format-ignores injections; Llama sits lowest overall; o4-mini solves near-perfectly and complies with 100% of injections. Benchmarks are not broken; treating one accuracy number as the complete picture is. Across 219 canonical problems (the full bank: 44 GSM, 65 BW, 110 ALGO), 6 variants, 5 models, and $\sim$ 20K model API calls (coverage in Table 2), three independent tests reveal patterns accuracy hides:

- Matched-benchmark models differ in W_3 robustness (GSM calibration at $n=44$ for Claude/Gemini/o4-mini, partial at $n=20$ for GPT-4o/Llama), and the direction reverses across problem types.
- Plan fidelity does not predict correctness: GPT-4o (median CCI=0) is correct $\sim$ 70% of the time.
- Accepting a wrong intermediate at 88–93% does not reliably derail the final answer.
- Same DP, different exposure: WIS collapses under W_3 universally—consistent with both lower exposure and higher baseline difficulty (Table 5); CC partially survives (o4-mini caveat, Section 4.3).

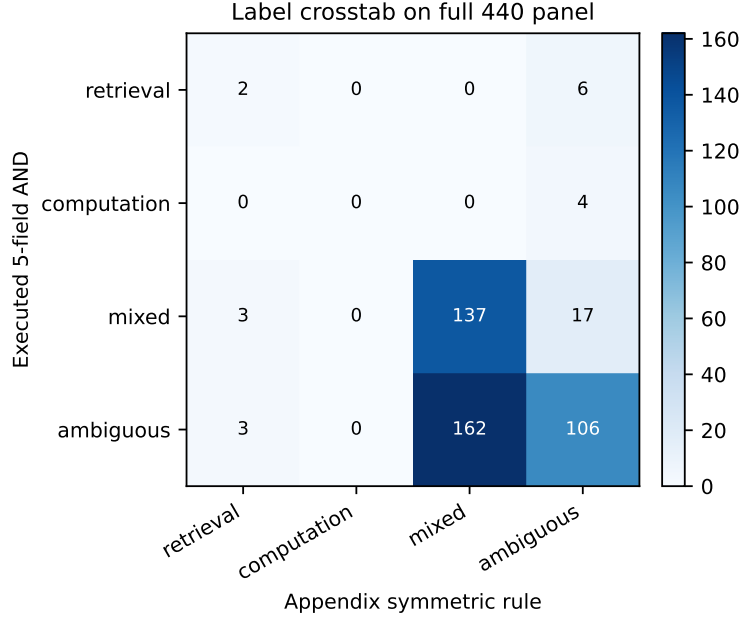


Figure 7: Crosstab of executed five-field AND vs. appendix symmetric three-signal labels on the full 440 panel (195/440 cells change).

- BW rename: Claude +23.9pp, Gemini +18.5pp, Llama -22.0pp; GPT-4o/o4-mini unchanged. Algorithm elicitation $\uparrow$ invocation 10–50 $\times$, accuracy unchanged (0/5, $\Delta=-2.0$ pp).

Report canonical and W_3 -rename accuracy with CIs—the gap is capability.

Limitations

Coverage, credits, and NOT_COMPUTABLE cells. All results are zero-shot CoT at $T=0$, English structured reasoning only. GSM GPT-4o/Llama Probe-1 covers $n=20$ bank-valid problems (GSM_001–020); Wilson CIs are ~ 30 pp wide—we report these as partial replication, not powered five-model evidence (headline inversions use ALGO SP-challenging, $n=34$). Under frozen filters, GSM W_6 for GPT-4o/Llama is NOT_COMPUTABLE (no bank-valid W_6 rows after ER-ROR filtering). Many ALGO CC/WIS W_5/W_6 cells are genuinely absent by design. o4-mini has no ALGO Phase-1 declaration file, so ALGO CCI for that model is NOT_COMPUTABLE. Several ALGO compliance Fishers are undefined because refusal $n=0$. Figure 8 summarises these holes. GSM_041–064 gap fill, remaining ALGO Phase-2A elicited sessions (49/110 for Claude/Gemini/Llama), BW Probe-2 expansion to Gemini/o4-mini, a difficulty-matched WIS bank, and second reasoning-trained providers (DeepSeek-R1 [Guo et al., 2025], Qwen3) are queued on restored API credits, not on study design. Section 4.3’s proximity-link observation is one-model-strong (o4-mini only). CCI comparisons across models partly reflect Phase-1 format propensity; 17/29 GPT-4o zero-CCI sessions are parser-empty by construction (Section 3.1). Few-shot demonstrations are known to partially restore accuracy under template perturbation [Mirzadeh et al., 2025]; we present W_3 numbers as zero-shot estimates, not capability upper bounds. WIS subtype collapse is observational: exposure and difficulty co-vary (Table 5). Idle analysis cells ($W_1/W_2/W_4$ bulk accuracy only) exist in the release bundle but are not claimed as primary results.

Measurement limits. The strict BW PDDL protocol aborts at 84–100%, yielding a protocol-level finding rather than per-model CCI. The NL-tolerant BW Probe-2 rerun covers *three* models (Claude, GPT-4o, Llama), not five—Gemini and o4-mini were not extended once the protocol was known-broken; earlier wording that said “all five” was incorrect. CCI is a behavioural metric only—it does not probe internal reasoning faithfulness. Infini-gram overlap is a public-corpus proximity proxy

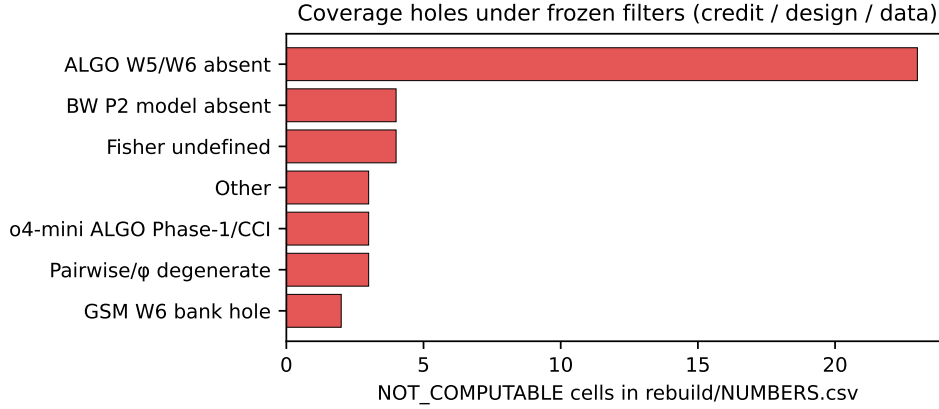


Figure 8: Counts of NOT_COMPUTABLE cells in the frozen number file, grouped by cause (variant absence, bank/API holes, missing Phase-1, undefined tests).

for closed-weight models, not a confirmed contamination signal; closed models have *no* calibratable corpus membership ground truth (permanent O16 caveat), whereas open models (Pythia/OLMo) can be checked against The Pile/Dolma. The within-ALGO CC→SP→WIS gradient is the cleaner internal control. Mechanistic pilots are Llama/Qwen-only and cannot support five-API-model claims. On triangulation, 127 parse failures and 196 missing Phase-2 rows force ambiguous labels on the full 440 panel; confident labels remain rare even after complete-case filtering. The full per-(probe, model) coverage matrix and extended scoping decisions appear in Appendix A.

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Table 6: Explicit per-(probe, model) coverage. All five models are fully run on Probe 1 and on GSM/ALGO Probe 2–3 except where footnoted. Strict-PDDL BW Probe-2 is n/a for Gemini and o4-mini (protocol aborts at 84–100% on the three models we ran); the NL-tolerant BW Probe-2 rerun covers those same three models only (Claude, GPT-4o, Llama). ^ao4-mini GSM Probe 2: 43/44 Phase-1 declarations parse; CCI/TEP computed on the parseable subset. ^bBW interactive PDDL not run on Gemini or o4-mini (protocol aborts at $\geq 84\%$ on the three models we did run); NL-tolerant BW Probe-2 likewise covers only those three models. ^cGPT-4o/Llama Probe-1 GSM: 20/44 bank-valid (GSM_001–020 only); GSM_041–064 pending API credit gap fill. ^do4-mini ALGO Phase-1 declaration file absent $\Rightarrow$ ALGO CCI NOT_COMPUTABLE.

	Claude	GPT-4o	Llama	Gemini	o4-mini
P1 GSM	44/44	20/44 ^c	20/44 ^c	44/44	44/44
P1 ALGO	110/110	110/110	110/110	110/110	110/110
P1 BW	65/65	65/65	65/65	65/65	65/65
P2 GSM (CCI/TEP)	44/44	44/44	44/44	44/44	43/44 ^a
P2A ALGO normal	110	110	110	110	110
P2A elicited	61	110	61	61	110
P2B plausible	61	61	61	61	61
P2B implausible	61	61	61	61	61
P2 BW (CCI)	50	50	50	n/a ^b	n/a ^b
P2 BW NL-tolerant	50	50	50	n/a ^b	n/a ^b
P3 contam. (VRI)	✓	✓	✓	✓	✓
P3 contam. (CCI)	✓	✓	✓	✓	n/a ^d
P3 mech.	Qwen/Llama pilots only (Appendix F)				

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377 A Coverage Matrix and Extended Limitations

378 **Explicit coverage matrix.** Table 6 provides the explicit per-(probe, model) coverage map; every
379 cell that is not a clean “run” is foot-noted with the reason (API error, unparseable-output, intentional
380 scoping). All quantitative claims in the main paper are scoped to fully-run (probe, model) cells;
381 o4-mini appears in main-paper figures and tables only where we have valid data, and is shown as
382 explicit n/a cells elsewhere.

Extended scoping decisions and future-work pointers.

1. **Probe-3 as proximity proxy by design.** Training-corpus ground truth is unavailable for closed-weight models. We present Infini-gram overlap on The Pile/DCLM as an explicitly labelled public-corpus proximity proxy and require three-signal convergence before applying a retrieval-consistent label.
2. **Matched-subset sample sizes are deliberate, not under-powered.** We report matched-canonical tests on the subsets where matching yields useful n (shortest-path $n=34$, coin-change $n=10$). The full-subtype ALGO CIs are cluster bootstrap over clone families (10k, seed 42); GSM/BW use Wilson. The matched tests are supporting evidence. The CC inversion ($n=10$) is explicitly labelled exploratory in the main text.
3. **WIS exposure-difficulty co-variance.** WIS is both lower exposure and harder—a known property of the algorithm class, not an oversight. We treat exposure as a co-occurring signal and flag the confound wherever the within-ALGO gradient is cited; a difficulty-matched WIS bank is a natural next experiment.
4. **CCI is a behavioural metric by construction.** CCI measures declared-vs-executed numeric agreement under hard session isolation; it deliberately does not probe internal reasoning, which we address separately via the Qwen-2.5-7B residual-stream pilot (Appendix F).
5. **Cross-tokeniser evidence is built into the design.** Claude (BPE-A) and GPT-4o (BPE-B) tokenise the same nonce vocabulary differently yet show opposite fragility patterns on the same problem set—this rules out tokenisation as the sole driver. A direct tokeniser ablation is a clean follow-up.
6. **Released artefacts.** The audit bundle ships per-instance verifier logs (the single source of truth) plus one legacy derived per-problem export, retained only for traceability; no main-paper table or figure depends on it.
7. **Reasoning-trained breadth.** We chose to run one reasoning-trained provider (o4-mini) at full coverage rather than two at partial coverage. Section 4.3’s proximity-link observation is therefore one-model-strong by design; replicating on DeepSeek-R1 [Guo et al., 2025] or Qwen3 is the targeted next sweep.
8. **Blocksworld protocol evolution.** The strict-PDDL protocol revealed a measurement-level finding: interactive PDDL evaluation is brittle at the API level. The NL-tolerant rerun canonicalises the BW Probe-2 result for the three models we ran (Claude, GPT-4o, Llama); Gemini and o4-mini were not extended after the protocol was known-broken. The strict-protocol abort taxonomy is reported as a separate methodological finding rather than as per-model behavioural CCI.
9. **Algorithm-invocation causality.** The five-model elicitation experiment establishes an associational null; proving the absence of causation requires controlled-prompt RCTs—a clean follow-up.
10. **Probe-2 BW coverage decision.** Once the strict-PDDL protocol aborted at $\geq 84\%$ on three models, the responsible call was not to extend a known-broken measurement to two more. The BW Probe-2 behavioural conclusion is driven by the NL-tolerant standard→mystery contrast on those three models (not five).
11. **API / credit backlog.** GSM_041–064 for GPT-4o/Llama, remaining ALGO Phase-2A elicited sessions, and optional BW Probe-2 expansion remain queued on restored credits (results/derived/PRE_API_MASTER_AUDIT.md).
12. **Closed-model contamination GT (O16).** Corpus membership ground truth is permanent-impossible for Claude/GPT-4o/Gemini/o4-mini; only open models admit Infini-gram GT calibration.
13. **Triangulation rule authority.** Published 8/4/157/271 counts come from the executed five-field AND; the appendix three-signal print is sensitivity-only (see rebuild/tri_v2/).

B Variant Generation Protocol

W1 (paraphrase). Problem statement rewritten in different words; all numeric values, entities, and structural relationships preserved exactly.

435 **W2 (format change).** Prose converted to structured table, edge list, interval table, or equation as
436 appropriate to the family (e.g., SP problems expressed as a weighted adjacency matrix).

437 **W3 (entity rename).** All entity identifiers replaced with nonce tokens drawn from a held-out vocabu-
438 lary: length-matched, tokenization-equivalent where feasible (single-token for single-token orig-
439 inals), and vocabulary-disjoint from the original. Claude (BPE-based) and GPT-4o (a different
440 BPE variant) show opposite W3 fragility directions on the same SP and CC subtypes, ruling out
441 a tokenizer-artifact explanation as the sole driver.

442 **W4 (formal notation).** LaTeX equations (GSM); PDDL-like formal predicates (BW); mathematical
443 edge-list notation (SP/WIS).

444 **W5 (direction reversal).** BW: goal and initial states swapped. GSM: forward problems inverted
445 to backward form (“what was the original amount?” instead of “how much remains?”). SP: source
446 and destination nodes swapped, testing whether the model re-solves from scratch or retrieves a
447 cached forward solution. W5 is not applied to CC or WIS (no meaningful reversal exists for those
448 optimization problems).

449 **W6 (procedural regeneration).** Same algorithm and parameter ranges; completely new numeric
450 values.

451 All variants pass the same domain-specific verifier as the canonical problem before any model is
452 queried (PDDL simulator for BW; exact-match for GSM; DP-optimal solver for ALGO).

453 C Metric Definitions

454 **W3 retention.** $R_{W_3} = \text{Acc}_{W_3} / \text{Acc}_{\text{can}}$; undefined where canonical accuracy is 0.

455 **VRI.** $\text{VRI} = [\text{Acc}_{W_1} + \text{Acc}_{W_2} + \text{Acc}_{W_4}] / 3 - \text{Acc}_{W_3}$.

456 **CCI (GSM).** $\text{CCI} = \frac{1}{k} \sum_{i=1}^k \mathbf{1}[|v_i^{(1)} - v_i^{(2)}| \leq 0.01]$. Sessions with $k=0$ are zero-imputed in between-
457 model paired tests.

458 **TEP (Trajectory Error Propagation).** Fraction of steps after the injection step whose numeric or
459 symbolic content differs from the uninjected run. Reported for GSM (from GSM Phase 2B) and
460 ALGO (from ALGO Phase 2B).

461 **Injection-compliance.** Classified at the injection step: compliant, partial, refusal, or format-ignored.
462 The four-way compliance taxonomy is applied to the ALGO Phase-2B injections (Section 4.2).

463 **Proximity.** Infini-gram [Liu et al., 2024] n -gram overlap with The Pile and DCLM, $[0, 1]$. Two
464 scores per problem: template contamination and instance contamination. In Table 1 this public-
465 corpus n -gram proximity is the P3-I component of Probe 3; P3-M denotes the complementary
466 mechanistic final-rank read-out (median gold-token rank in the residual stream), reported only for
467 the open-weight Qwen-2.5-7B pilot (Appendix F).

468 **Statistics.** Wilson 95% CIs for GSM and BW proportions. ALGO intervals are 10k percentile cluster
469 bootstrap over clone families (seed 42), not Wilson and not iid problem bootstrap. Fisher’s exact
470 on matched-canonical subsets (exploratory only). Paired Wilcoxon signed-rank (primary) and paired
471 t -test (secondary) for CCI/TEP comparisons ($n=44$, zero-imputed). For continuous proximity-
472 behavior correlations we report Pearson r as primary; Spearman ρ as a robustness check (same
473 conclusions). Bootstrap p -values from 10,000 resamples for all correlations. **Multiple-comparison**
474 **policy.** We pre-specified two confirmatory hypothesis families (the three Probe-1 family-level reten-
475 tion findings; the five paired-Wilcoxon plausibility tests in Probe-2B) and report Holm–Bonferroni
476 adjusted p -values for both: every primary test that survives unadjusted ($p < 0.05$) also survives Holm–
477 Bonferroni at $\alpha=0.05$; the borderline ALGO ranking test ($p=0.04$, $n=5$) does not, and is presented
478 as directional throughout.

479 D Full Per-Variant Accuracy and Cross-Probe Correlation

480 **BW subset note.** BW Probe 1 accuracies are computed from `behavioral_correct` restricted
481 to the $n=65$ question-bank IDs (50 standard Blocksworld + 15 Mystery Blocksworld), drawn from
482 PlanBench [Valmeekam et al., 2023b]. The shipped raw file `BW_P1_behavioral.csv` additionally

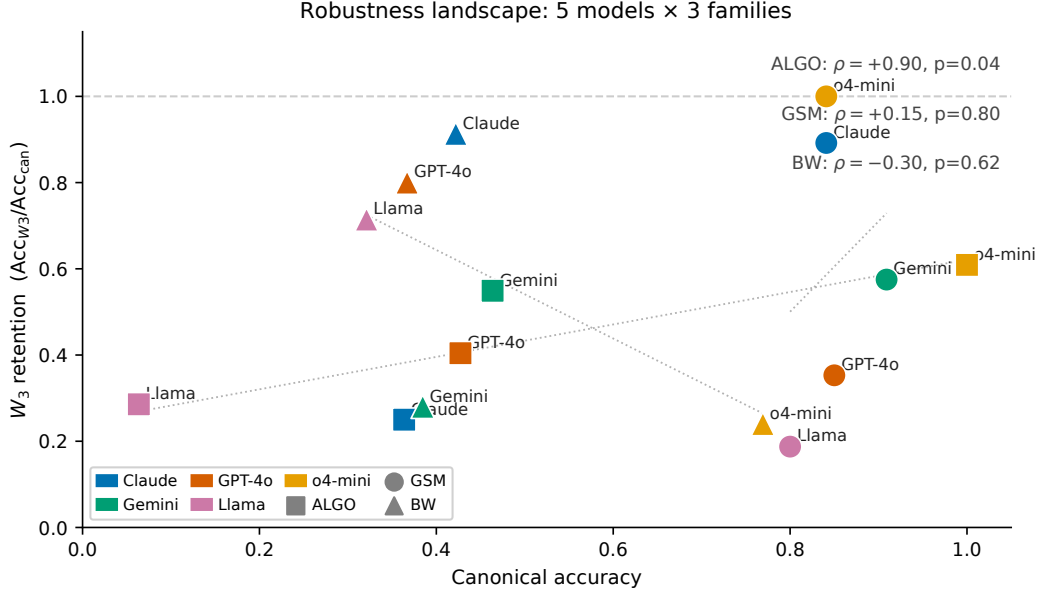


Figure 9: Accuracy–robustness landscape across five models and three families. Each point is one (model, family) pair. Within ALGO, more accurate models retain more of their canonical accuracy under W_3 ($\rho=+0.90$, $p=0.04$); the GSM trend is the same direction but n.s. ($\rho=+0.15$, $p=0.80$). On BW the rank-order is essentially flat ($\rho=-0.30$, $p=0.62$). o4-mini leads on GSM and ALGO retention but sits near the bottom on BW—a family-specific fragility invisible from absolute accuracy.

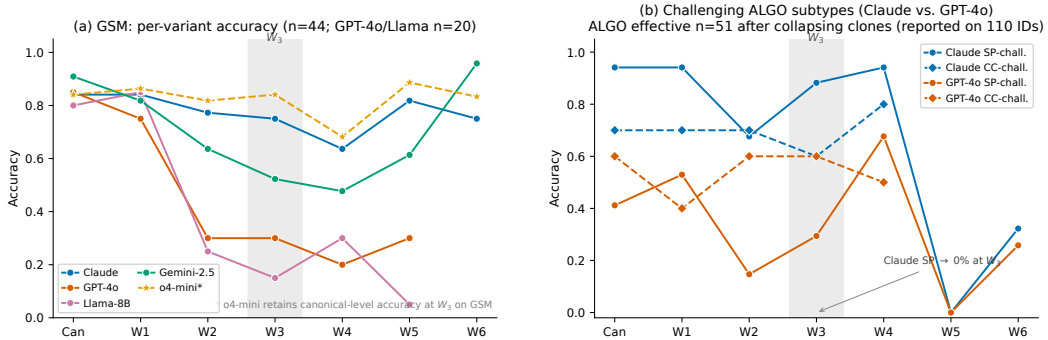


Figure 10: Per-variant accuracy profiles (Probe 1). (a) GSM: W_3 drops sharply for GPT-4o, Llama, and Gemini; for Claude it is mid-ranked. W_6 (new numbers) stays much closer to canonical than W_3 for the higher-accuracy models. (b) Challenging ALGO subtypes (Claude vs. GPT-4o): the dissociation at W_3 reverses between SP and CC.

contains 44 PlanBench-extended instances; if those are included unfiltered, canonical accuracies rise to 0.42 (Claude), 0.37 (GPT-4o), and 0.32 (Llama), because the extra problems are markedly easier than the bank subset. We report the bank-restricted numbers throughout for cross-family comparability.

GSM source files. GSM Probe-1 numbers in Tables 3 and 7 are computed from the per-model raw files (GSM_P1_behavioral_{claude,gpt4o,llama,gemini,o4mini}.csv), with $n=44$ for Claude/Gemini/o4-mini and $n=20$ for GPT-4o/Llama (GSM_001–020 only; GSM_021–040 are off-bank duplicate reruns and GSM_041–064 are pending API gap fill, both excluded).

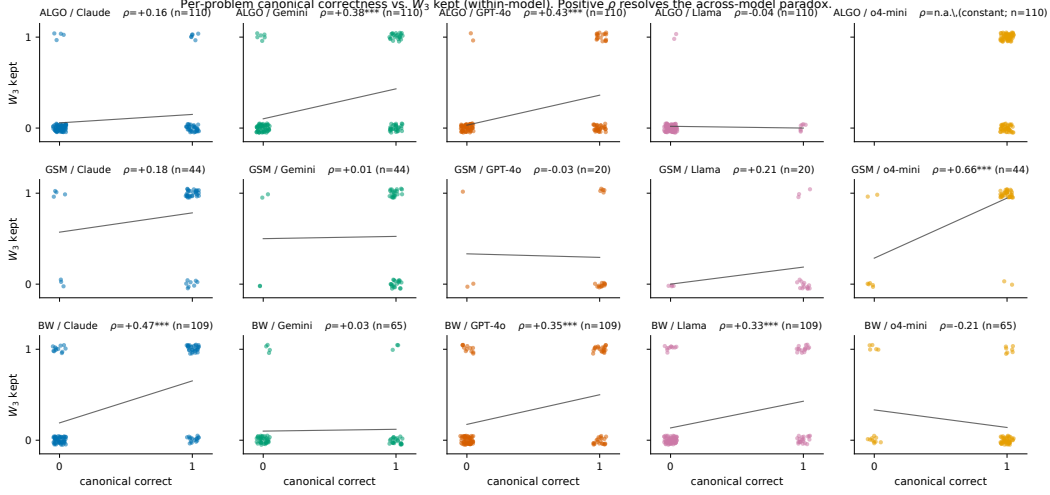


Figure 11: Within-model per-problem canonical correctness vs. W_3 correctness (3 families $\times$ 5 models). Each cell reports the Spearman ρ between two binary indicators (** $p < 0.01$, *** $p < 0.001$, * $p < 0.05$). Positive ρ in most cells resolves the model-rank-level finding into a within-model rule: harder canonical problems also lose more under W_3 . Llama and o4-mini on BW are the only flat/negative cells.

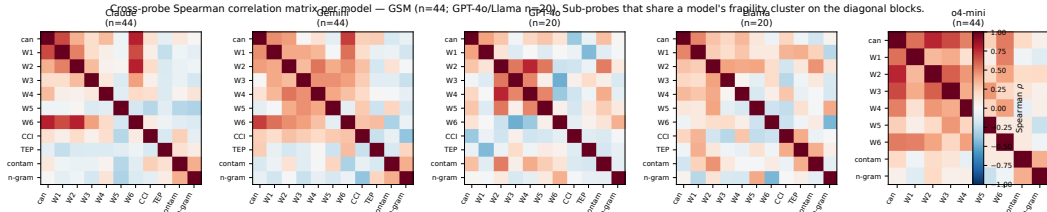


Figure 12: (a) GSM macro metrics: Gemini peaks on Acc_{can} (0.91) and CCI (0.27) but only mid-pack on R_{W_3} (0.57); o4-mini reaches $R_{W_3} = 1.00$. (b) Proximity predicts VR1 for Claude ($r = +0.44^{***}$) and GPT-4o ($r = +0.37^{**}$) only; predicts CCI for Claude ($r = +0.31^*$).

491 **o4-mini scoring notes.** o4-mini Blocksworld results in Table 7 ($n = 65$ bank, canonical .769)
 492 are computed from the NL-tolerant rerun; the original PDDL-strict pass scored 0/65 because PDDL
 493 tuples do not match the NL grader applied to other models. ALGO o4-mini accuracy is computed
 494 from the per-instance verified column (canonical 110/110 on the full instance pool).

495 **Population-level dissociation test.** We test whether the pairwise inversions in Section 4.1
 496 generalise to a population-level inverse relationship between canonical accuracy and W_3 re-
 497 tention (Figure 15). At the subtype $\times$ model cell level, pooled across GSM, ALGO and BW:
 498 Spearman $r = +0.147$ ($p = 0.46$, $n = 28$, 95% bootstrap CI $[-0.27, +0.56]$); ALGO-only $r = +0.37$
 499 ($p = 0.10$, $n = 20$). At the per-problem level the phi coefficients within each model are pre-
 500 dominantly non-negative: ALGO $\phi_{\text{Claude}} = +0.16$, $\phi_{\text{Gemini}} = +0.38$, $\phi_{\text{GPT-4o}} = +0.43$, $\phi_{\text{Llama}} = -0.04$;
 501 GSM $\phi_{\text{GPT-4o}} = -0.03$ ($n = 20$ bank-valid), $\phi_{\text{o4-mini}} = +0.66$; BW $\phi_{\text{Claude}} = +0.47$, $\phi_{\text{GPT-4o}} = +0.35$,
 502 $\phi_{\text{Llama}} = +0.33$. The only clearly negative cells are Llama on ALGO and o4-mini on BW. The
 503 aggregate Spearman is indistinguishable from zero, so the pairwise inversions are real per-(model,
 504 subtype) phenomena, not a universal inverse axis.

505 E Probe 2 Protocol Details

506 **GSM Probe 2.** *Phase 1 (Declaration).* The model produces a complete step-by-step solution with
 507 an explicit intermediate numeric value at each step (step index, operation, numeric output). *Phase*
 508 *2A (Normal execution).* A fresh API session per step; the model receives the original problem, step

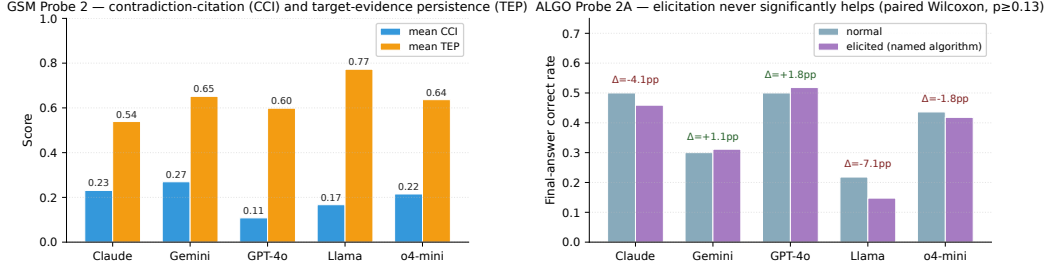


Figure 13: Probe-2 summary across five models. *Left*: GSM contradiction-citation index (CCI) and target-evidence persistence (TEP) per model on the full 44-problem GSM-P2 corpus. CCI rarely exceeds 0.30 for any model; TEP is high for every model (mean ~ 0.6 – 0.8): the injected wrong value propagates through many subsequent steps, yet the final answer survives at near-baseline rates. *Right*: ALGO Probe-2A final-answer accuracy with and without algorithm-elicitation; per-model deltas range from -7.1pp (Llama) to $+1.8\text{pp}$ (GPT-4o), and the pooled paired Wilcoxon over $n=403$ problem-instances is $p=0.20$. Elicitation never significantly improves final accuracy.

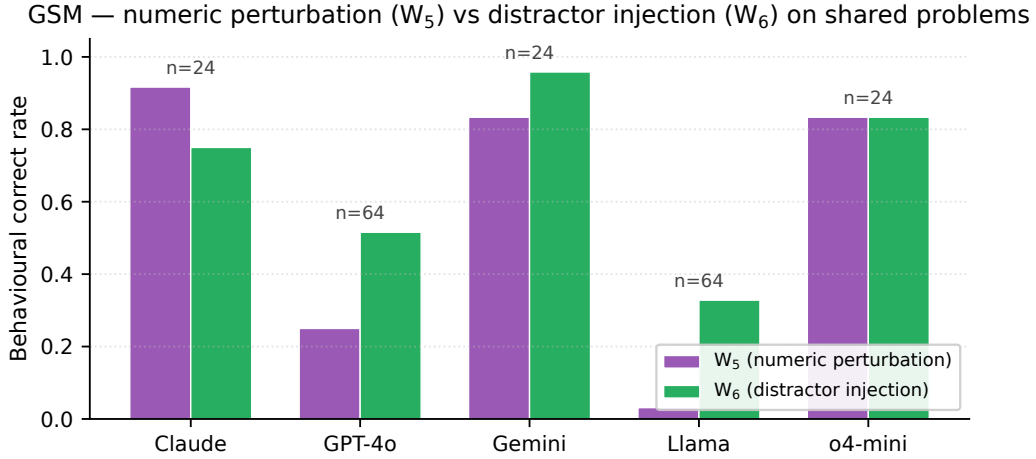


Figure 14: GSM numeric perturbation (W_5) vs. distractor injection (W_6) on shared problems, per model. W_6 (new numbers, same algorithm) stays close to canonical accuracy for the higher-accuracy models, while W_5 (direction reversal) produces larger drops. Sample sizes vary by model due to the bank-valid subset constraint ($n=24$ for Claude/Gemini/o4-mini; $n=40$ for GPT-4o/Llama).

index, and minimal state—no Phase-1 plan and no chat history from earlier Phase-2A steps. Each Phase-2A output is compared to the Phase-1 declared value via $\text{match}_i = \mathbf{1}[|v_i^{(1)} - v_i^{(2)}| \leq 0.01]$; CCI is the mean. *Phase 2B (Injection)*. Identical to 2A up to a uniformly sampled injection step (excluding the first and last 10% of the trajectory); the true intermediate value is replaced with a plausible-magnitude but incorrect value. Subsequent steps are classified for TEP relative to the uninjected run.

ALGO Probe 2. *Phase 1.* Four-field JSON prompt: (1) named algorithm and why; (2) will greedy work, why/why not; (3) first decision from the initial state; (4) at which step greedy is most likely to fail. *Phase 2A.* Fresh session per step with problem, current state, and minimal step context (no prior chat history); 30-step budget. *Phase 2B.* As 2A, except the injection step replaces the true state with one frontier-cost modification (SP), one interval-boundary modification (WIS), or one coin-denomination addition (CC), designed to be locally plausible but globally incorrect if propagated.

Plausible/implausible-injection aggregate gap. In the post-injection arm, the aggregate “implausible” condition has a higher correct rate (54.1%, $n=122$) than “plausible” (39.3%, $n=244$; Fisher $p=0.010$). This aggregate gap is a sampling artifact: the implausible arm contains only Claude and

Table 7: Per-variant accuracy under frozen rebuild filters (rebuild/NUMBERS.csv, rebuild/FROZEN_FILTERS.md). GSM bank-valid $n=44$ (GPT-4o/Llama Probe-1 currently $n=20$ on GSM_001-020; W6 for those models is NOT_COMPUTABLE after bank+ERROR filter). ALGO slices use the frozen adversarial/standard split (SP+CC+WIS challenging $n=61$). BW bank $n=65$ PlanBench IDs. Dashes mark absent variants or non-computable cells (API/credit gaps or design holes), not zero accuracy.

Family	Slice	Model	Can.	W1	W2	W3	W4	W5	W6
GSM	–	Claude	.841	.841	.773	.750	.636	.818	.750
GSM	–	GPT-4o	.850	.750	.300	.300	.200	.300	—
GSM	–	Gemini	.909	.818	.636	.523	.477	.614	.958
GSM	–	Llama	.800	.850	.250	.150	.300	.050	—
GSM	–	o4-mini	.841	.864	.818	.841	.682	.886	.833
ALGO	CC-chall	Claude	.700	.700	.700	.600	.800	—	—
ALGO	CC-chall	GPT-4o	.600	.400	.600	.000	.500	—	.400
ALGO	CC-chall	Gemini	.500	.700	.600	.700	.700	—	—
ALGO	CC-chall	Llama	.200	.100	.400	.000	.200	—	.100
ALGO	CC-std	Claude	.267	.467	.067	.200	.667	—	.067
ALGO	CC-std	GPT-4o	.267	.400	.000	.067	.867	—	.200
ALGO	CC-std	Gemini	.267	.133	.000	.000	.267	—	.267
ALGO	CC-std	Llama	.000	.067	.067	.000	.000	—	.067
ALGO	SP-chall	Claude	.647	.618	.676	.000	.824	.000	.258
ALGO	SP-chall	GPT-4o	.412	.529	.147	.265	.588	.000	.265
ALGO	SP-chall	Gemini	.676	.441	.235	.324	.559	.032	.129
ALGO	SP-chall	Llama	.059	.147	.029	.000	.088	.000	.059
ALGO	SP-std	Claude	.000	.190	.667	.048	.952	.000	.000
ALGO	SP-std	GPT-4o	.714	.667	.048	.429	.524	.000	.381
ALGO	SP-std	Gemini	.619	.762	.762	.476	.857	.000	.263
ALGO	SP-std	Llama	.048	.095	.000	.000	.143	.000	.095
ALGO	WIS-chall	Claude	.353	.176	.118	.000	.059	—	.000
ALGO	WIS-chall	GPT-4o	.353	.176	.000	.000	.000	—	.000
ALGO	WIS-chall	Gemini	.353	.176	.000	.000	.000	—	.000
ALGO	WIS-chall	Llama	.059	.000	.000	.059	.000	—	.000
ALGO	WIS-std	Claude	.077	.231	.231	.000	.077	—	.000
ALGO	WIS-std	GPT-4o	.154	.231	.000	.000	.000	—	.000
ALGO	WIS-std	Gemini	.000	.000	.231	.000	.000	—	.000
ALGO	WIS-std	Llama	.000	.000	.000	.077	.000	—	.000
BW	–	Claude	.154	.062	.231	.138	.015	.523	.508
BW	–	GPT-4o	.062	.092	.092	.169	.077	.246	.215
BW	–	Gemini	.385	.138	.108	.108	.031	.569	.338
BW	–	Llama	.015	.031	.015	.108	.000	.000	.031
BW	–	o4-mini	.769	.754	.738	.185	.415	.769	.769

GPT-4o sessions, while the plausible arm additionally contains Llama (22.9%) and Gemini (31.1%), which depress the plausible mean. Within the two models tested on both arms, the difference is 0pp for Claude (52.5% in both) and +4.9pp for GPT-4o (50.8% vs. 55.7%). We therefore draw no behavioral conclusion from the aggregate gap.

Reasoning-type classification. Each Phase-2A step is assigned one of five reasoning types—unclear, local-greedy, forward-simulation, algorithm-invocation, or backtracking—by a deterministic rule-based classifier that matches lexical and structural cues in the step text (e.g. an explicit named algorithm such as “Dijkstra” or “dynamic programming” for algorithm-invocation; an explicit revision of a prior decision for backtracking). The decision rules are listed in Appendix K. Because the classifier is deterministic, its labels are exactly reproducible from the released step logs; a complementary LM-judge cross-check on a 100-step sample is a planned extension.

Blocksworld Probe 2. *Phase 1.* The model declares a full sequence of PDDL (action object) tuples. *Phase 2A.* Each declared step is executed within a PDDL state-transition simulator; legal actions advance the state, illegal actions are recorded, and sessions abort after 15 consecutive illegal actions. 84–100% of sessions abort before plan completion (Section 4.4).

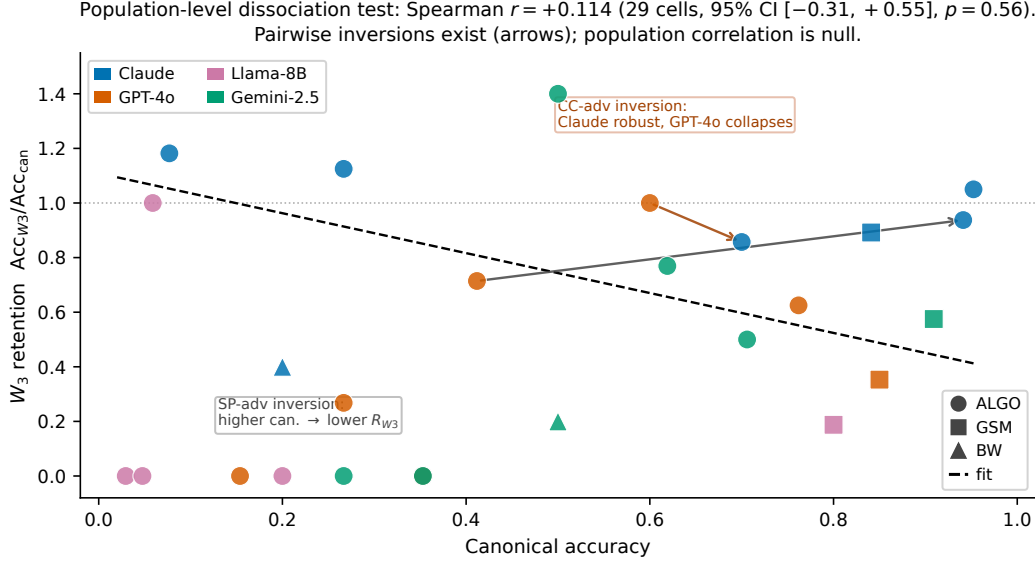


Figure 15: Population-level dissociation test. Each point is one (model, family/subtype, instance-type) cell. The two matched-canonical inversion pairs that drive the headline finding (SP-adv and CC-adv, Claude $\leftrightarrow$ GPT-4o) are highlighted. Pooled Spearman is positive but indistinguishable from zero, so the dissociation is genuinely per-(model, subtype), not a universal axis.

F Mechanistic Rank Dissociation (Qwen pilot)

Why we ran a mechanistic check. A behavioural W_3 collapse alone cannot distinguish “model genuinely recomputes but fails surface mapping” from “model surfaces a memorised canonical answer that no longer matches the renamed problem.” The two read-outs are confounded behaviourally but make opposite predictions inside the residual stream of an open-weight model: under retrieval-style commitment, the canonical gold token should remain highly accessible at the final layer even though the problem has been renumbered; under recomputation, those signatures should shift toward the variant’s gold token. We run an open-weight pilot on Qwen-2.5-7B [Qwen Team, 2024] (28 transformer layers), reading out four mechanistic signatures of the canonical gold token per (problem family, variant): (1) terminal commitment (median final-layer rank); (2) depth-resolved rank trajectory; (3) depth-resolved calibrated confidence ($\log p(\text{gold} \mid \text{prompt})$); and (4) depth-resolved representational alignment (cosine similarity to the gold unembedding direction). Llama-3.1-8B [Dubey et al., 2024] is the priority follow-up.

What the four metrics show. *Terminal commitment.* Algorithmic: canonical median rank 22,472 collapses to 156 under W6 ($n_{\text{can}}=100$, $n_{W_6}=90$; $\sim 144\times$ more accessible). Planning: 419 $\rightarrow$ 49 ($n=65$; $\sim 9\times$). Arithmetic: 2,985 $\rightarrow$ 3,201 ($n=44/24$; a reversal). *Depth-resolved rank.* For algorithmic and planning, canonical and W6 trajectories are indistinguishable through layer ~ 24 then split sharply in the last 3–4 layers; for arithmetic both collapse together. *Log-prob* and *cosine similarity* show the same dissociation, with W6 keeping the residual stream closer to the gold direction from layer ~ 5 onward.

Paired statistics on aligned pairs. One-sided Wilcoxon (H_1 : canonical rank $>$ W6 rank): planning $n=65$, median $\log_{10}(\text{can}/W_6)=0.80$, $W=1359$, $p=2.4\times 10^{-6}$; algorithmic $n=90$, median $=0.00$, $W=1072$, $p=0.079$ (borderline); arithmetic $n=24$, median $=-0.39$, $p=0.63$ (reversal confirmed). The “W6 surfaces gold token” claim is statistically robust for planning, directional but not population-significant for algorithmic, and reversed for arithmetic.

The four metrics are not redundant. Final-token rank and log-probability correlate strongly ($r=-0.79$), but final rank and final cosine similarity correlate only weakly ($r=-0.17$, $p<10^{-3}$)

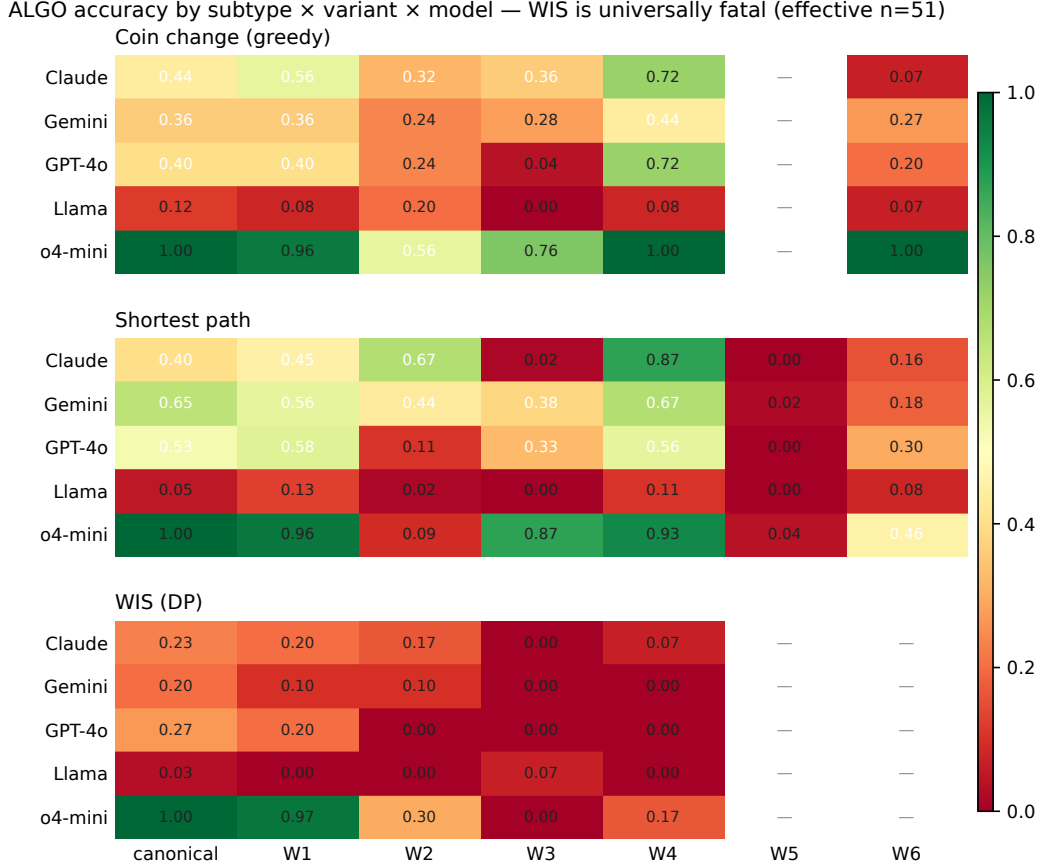


Figure 16: ALGO accuracy decomposed by subtype × variant × model (3 subtypes × 7 variants × 5 models). The WIS row is the universal Achilles heel of every model; shortest-path W5 is a hard floor across the suite; coin change (CC) is the only subtype where renaming retention is non-trivial. (Note: CC is a dynamic-programming problem; the point of the subtype is precisely that a greedy heuristic is *not* optimal for the general coin systems used here.) o4-mini’s perfect canonical row does not carry to W3/W6 on WIS, isolating algorithmic depth as a distinct robustness axis.

566 and final log-probability and final cosine similarity are statistically independent ($r=+0.005$, n.s.).
 567 We present all four because each isolates a distinct aspect of representational alignment.

568 **Scope and limitations.** This is a single open-weight backbone; we do not claim the same per-
 569 layer dynamics hold for frontier closed-weight models. The mechanistic result is supportive of the
 570 behavioural pattern, not an explanation of it.

571 G Evaluation Pipeline Design

572 **Data flow.** Four sequential stages: (1) **Canonical problem bank.** GSM: stratified sample from
 573 GSM8K [Cobbe et al., 2021] across four Infini-gram contamination quartiles. ALGO: hand-
 574 designed CC/SP/WIS instances with verified DP-optimal solutions. BW: instances from Plan-
 575 Bench [Valmeekam et al., 2023b]. (2) **Variant generation.** Each canonical problem expanded
 576 to 6 variants by deterministic scripts; every variant passed through a domain verifier before storage.
 577 (3) **Model sweep.** All (problem, variant, model) triples queued via OpenRouter at $T=0$, zero-shot
 578 CoT; Phase-2 probes run in strict session isolation. (4) **Verification and scoring.** Family-specific
 579 verifiers; per-instance outputs written to raw CSVs that are the single source of truth.

Plausible vs. implausible injection: no model differentiates (ALGO Probe 2B, 5 models, paired Wilcoxon)

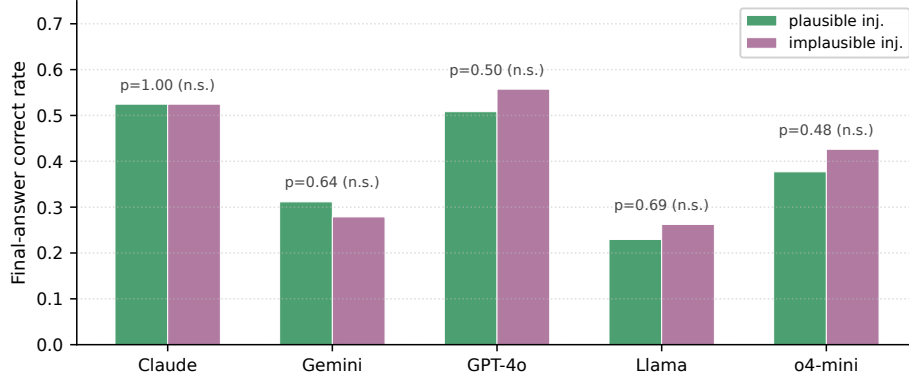


Figure 17: Plausible vs. implausible mid-stream injection, ALGO Probe 2B, $n=61$ paired problems per model. Paired-Wilcoxon p -values shown above each pair; no model differentiates the two arms.

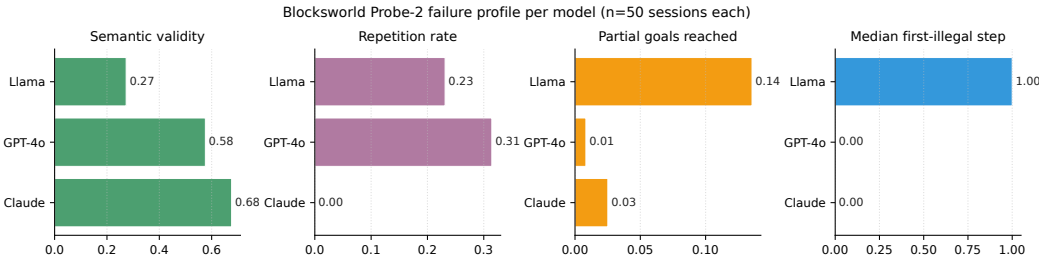


Figure 18: Blocksworld Probe-2 failure profile per model ($n=50$ sessions each). Claude, GPT-4o, and Llama reach near-identical low CCI through three distinct breakdown modes: parse-formattable but semantically valid plans (Claude), high format errors with retained semantic validity (GPT-4o), and broad collapse plus heavy action repetition (Llama).

580 H Reproducibility and Verification

581 All code, per-instance verifier logs, and variant generation templates are at <https://anonymous.4open.science/r/retrieval-vs-computation-EA8F/README.md> and will be moved
582 to a permanent home upon acceptance. The bundle ships raw per-instance verifier logs with a re-
583 derivation script that recomputes every aggregate cited in the main paper directly from those logs;
584 the script applies the bank-ID filter so GSM GPT-4o/Llama figures reproduce at $n=20$. Reviewers
585 can verify reported numbers without retraining or re-querying any model. The repository contains
586 no author-identifying information. **Wilcoxon sensitivity.** Full bank $n=44$ (zero-imputed): $W=396$,
587 $p=0.0068$; paired $t=2.63$, $p=0.0072$. Complete-case analysis yields the same p -values.

589 I Algorithm-Invocation Observation (Exploratory)

590 Thirteen Phase-2 steps across Claude (6), Llama (4), and Gemini (3) were classified as *algorithm-*
591 *invocation*: the model explicitly named a specific algorithm (“I will use Dijkstra’s algorithm”) and
592 attempted to apply it. All 13 yielded incorrect final answers (Figure 22). Wilson 95% CI on 0/13:
593 $[0\%, 22.8\%]$. Fisher’s exact vs. unclear-reasoning baseline ($138/1,041=13.3\%$): $p=0.40$. This is
594 an observational pattern with $n=13$, not a statistical claim. Supporting observations: (a) distributed
595 across three of four models (GPT-4o produced zero algorithm-invocation steps across the entire
596 ALGO sweep); (b) selection effect ruled out (the 12 unique problems with algorithm-invocation
597 steps have *higher* mean canonical accuracy, 0.292, than the full ALGO set, 0.239; $t=0.81$, $p=0.42$);
598 (c) consistent qualitative pattern (the model named the correct algorithm, then executed a heuristic
599 inconsistent with it). Table 8 lists 10 of the 13 documented cases (3 Gemini cases omitted for space).

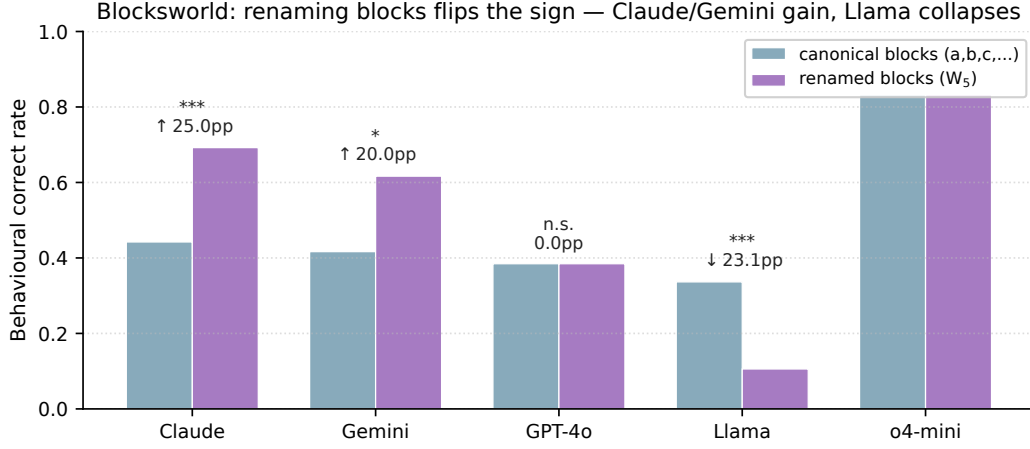


Figure 19: Blocksworld canonical vs. renamed-blocks (W5) accuracy, per-model paired Wilcoxon. Renaming the canonical a,b,c,... block tokens *improves* Claude/Gemini (removes harmful retrieval cues), *destroys* Llama (removes load-bearing cues), and leaves GPT-4o/o4-mini effectively untouched. The opposite signs across architectures are not consistent with a single mechanism.

Mechanistic dissociation (Qwen-2.5-7B residual stream): W₆ surfaces the gold token across all four depth-resolved metrics for algorithmic and planning families, but not for arithmetic

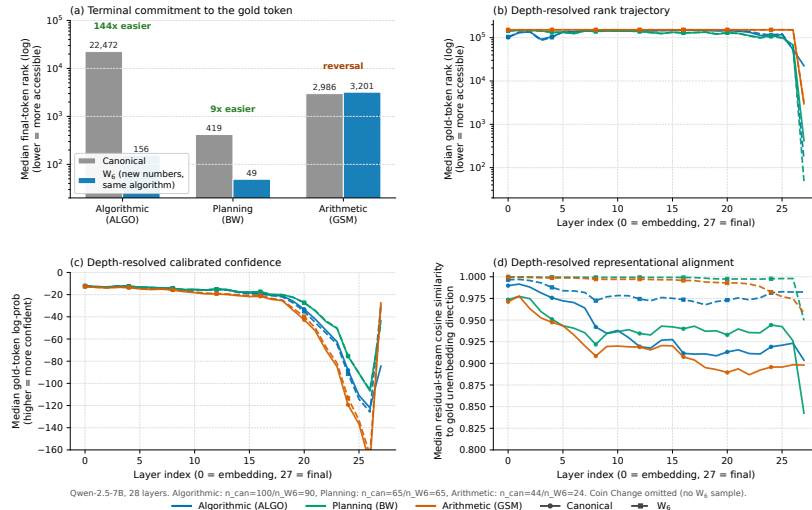


Figure 20: Mechanistic dissociation in the Qwen-2.5-7B residual stream, canonical vs. W₆ (new numbers, same algorithm), across three families. (a) Median final-token rank of the gold answer (log y -axis, lower is more accessible). (b)–(d) Layer-by-layer median trajectories of gold-token rank, log-probability, and residual-stream cosine similarity. Algorithmic and planning families show the predicted dissociation on all four metrics; the arithmetic family does not (the final-layer rank moves in the opposite direction), so the mechanistic claim is scoped to algorithmic and planning, not arithmetic.

600 J Triangulation Convergence Detail

601 ALGO (440 instances = 110 problems $\times$ 4 models; o4-mini excluded from the primary
602 panel because its near-ceiling W3 on several ALGO slices makes retrieval-vs-computation labels
603 poorly powered): under the *executed* asymmetric five-field AND rule the counts are retrieval=8,
604 computation=4, mixed=157, ambiguous=271 (61.6%). On the complete-case subset ($n=169$;
605 W3+CCI+proximity present and parse OK) ambiguous drops to 0 while 8/4/157 remain. GSM
606 (Claude, $n=44$): converging-computation=35; diverging=9. The appendix-style symmetric three-

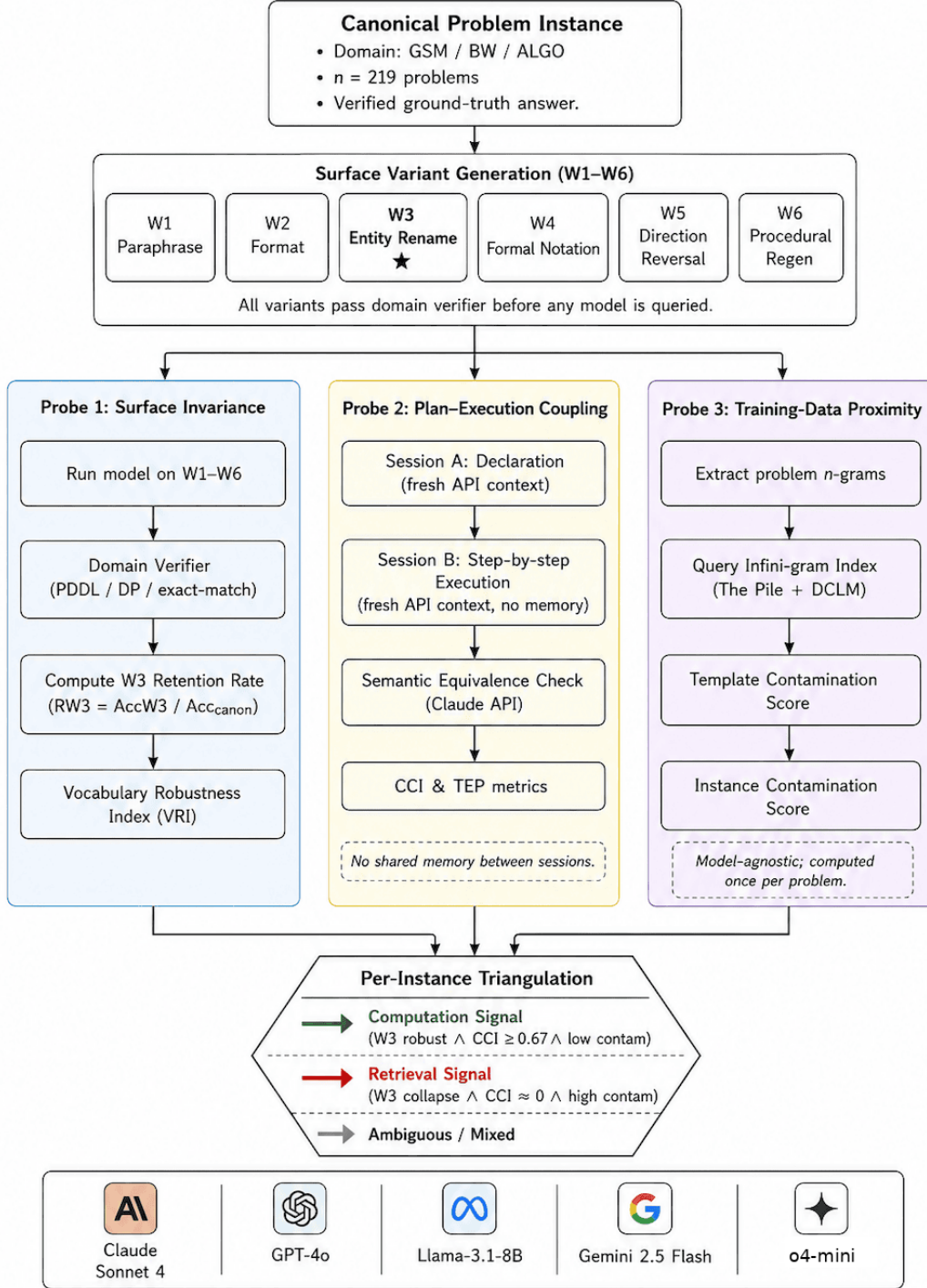


Figure 21: End-to-end pipeline. Each canonical problem generates six surface variants (W1–W6); Probe 1 records per-variant accuracy; Probe 2 spans three phases (declaration, execution, injection); Probe 3 queries the Infini-gram public-corpus index. All five models (Claude Sonnet 4, GPT-4o, Llama-3.1-8B, Gemini 2.5 Flash, o4-mini) are evaluated on every probe; the per-(probe, model) coverage matrix with intentional-scoping cells is in Table 6.

607 signal print (W3 at 0 or 1; $CCI \leq 0.10$ or ≥ 0.67 ; contamination at floor or above the 75th per-

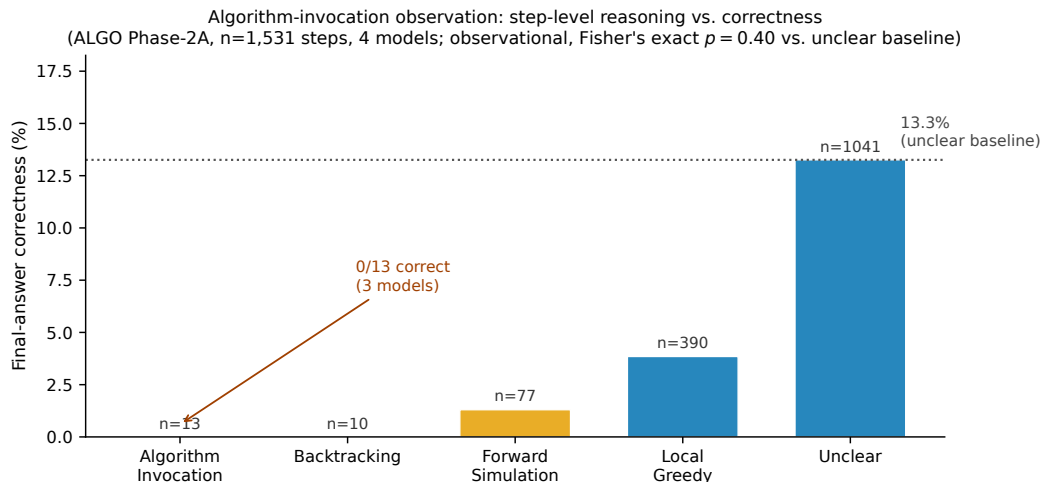


Figure 22: Step-level reasoning type vs. final-answer correctness across ALGO Phase-2A steps (4 models). Algorithm-invocation ($n=13$ steps, three models; $n=12$ unique problems—one problem contributed two invocation steps) yields 0% correct final answers. Dotted line shows the unclear-reasoning baseline (13.3%). Fisher’s exact: $p=0.40$ —not significant. This is an observational pattern, not a statistical claim.

Table 8: Algorithm-invocation cases (10 of 13; 3 Gemini cases omitted for space). All yielded incorrect final answers; in every case Phase-2A execution reverted to a greedy heuristic after naming the algorithm.

Case	Problem	Slice	Model
A1	SP_002	SP-std	Llama
A2	SP_004	SP-chall	Claude
A3	SP_037	SP-chall	Claude
A4	SP_040	SP-chall	Claude
A5	SP_044	SP-chall	Llama
A6	SP_062	SP-chall	Claude
A7	SP_065	SP-chall	Claude
A8	WIS_003	WIS-chall	Llama
A9	WIS_007	WIS-std	Llama
A10	WIS_020	WIS-chall	Claude

centile) is a *sensitivity analysis*: on the same 440 panel it yields 8/0/299/133 and disagrees with the executed rule on 195/440 instances. A 270-config threshold sweep collapses to 162 distinct configurations once identical binary-W3 cutoffs are merged; only 8 unique label vectors appear empirically (rebuild/tri_v2/).

K Prompts

Probe 1 (canonical and variants, all families).

Solve the following problem exactly and provide only the final answer in the required output format. Problem: {problem}. Format instruction: {family_specific_output_format}. The same template is used for canonical and W1-W6; only {problem} changes.

ALGO Phase 1 (declaration).

You are given the following problem: {problem}. Before solving it, answer four structured questions: (1) Which named algorithm will you use, and why is it appropriate? (2) Would a greedy approach work? Why or why not? (3) What is your first decision from the initial state? (4) At which step is

Table 9: Triangulation label rates on 440 ALGO instances under strict vs. liberal threshold configurations. The strong-label rate is a design choice, not a fixed property of the data.

Configuration	Retrieval	Computation	Strong total	Ambiguous
Legacy strict (paper default)	1.8%	0.9%	2.7%	61.6%
Liberal v2 sweep (param 204)	27.3%	30.4%	57.7%	37.9%

623 the greedy approach most likely to fail? Respond with a JSON object using
624 exactly these keys: {algorithm, greedy_works, first_move, critical_step}.

625 **ALGO Phase 2A (step execution).**

626 You are solving the following problem: {problem}. Current state: {state}.
627 Your prior decisions: {trajectory}. What is your next decision? Provide
628 a brief justification (1-2 sentences).

629 **GSM Phase 1 (numeric plan declaration).**

630 Solve the following arithmetic problem step by step. Before computing
631 each step, state what you will compute and why. After each step, write the
632 intermediate numeric result explicitly on its own line in the format: Step
633 N result: [value]. Problem: {problem}.

634 **Reasoning-type classification (deterministic rules).** Each Phase-2A step is assigned exactly one
635 category by a rule-based classifier over the step text, with the following decision criteria (evaluated
636 in order; default unclear):

- 637 • algorithm_invocation — the step names a specific algorithm (e.g. “Dijkstra”, “dy-
638 namic programming”, “knapsack”) and attempts to apply it.
- 639 • backtracking — the step explicitly revises or undoes a prior decision.
- 640 • forward_simulation — the step explicitly considers future states or look-ahead paths.
- 641 • local_greedy — the step selects the option with the lowest immediate cost without con-
642 sidering future states.
- 643 • unclear — none of the above cues are present.

644 **Probe 3 (training-data proximity query template).**

645 For each problem text, compute template-level and instance-level n -gram
646 overlap against indexed corpora (The Pile, DCLM) and return a normalized
647 score in [0,1] for each field.

648 Verifier outputs use deterministic parsing and exact checks: numeric exact-match (GSM), DP-
649 optimal answer match (ALGO), and goal-state reachability (BW; strict and NL-tolerant variants
650 both reported).

651 **NeurIPS Paper Checklist**

652 **1. Claims**

653 Question: Do the main claims made in the abstract and introduction accurately reflect the
654 paper’s contributions and scope?

655 Answer: [Yes]

656 Justification: Abstract and Section 1 state the three-probe diagnostic claim, matched-subset
657 inversions, and coverage limits; Limitations scopes partial GSM $n=20$ and API/credit
658 holes.

659 **2. Limitations**

660 Question: Does the paper discuss the limitations of the work performed by the authors?

661 Answer: [Yes]

662 Justification: Section 5 and Appendix A document API/credit backlog,
663 NOT_COMPUTABLE cells, NL-tolerant BW three-model coverage, O16 closed-model
664 GT, and triangulation missingness.

665 **3. Theory assumptions and proofs**

666 Question: For each theoretical result, does the paper provide the full set of assumptions and
667 a complete (and correct) proof?

668 Answer: [N/A]

669 Justification: This is an empirical evaluations paper; no theorems are claimed.

670 **4. Experimental result reproducibility**

671 Question: Does the paper fully disclose all the information needed to reproduce the main
672 experimental results of the paper to the extent that it affects the main claims and/or conclu-
673 sions of the paper (regardless of whether the code and data are provided or not)?

674 Answer: [Yes]

675 Justification: Frozen filters, number file, prompts, and verifier rules are in Appendix K/A
676 and `rebuild/FROZEN_FILTERS.md`; raw logs live under `results/raw/`.

677 **5. Open access to data and code**

678 Question: Does the paper provide open access to the data and code, with sufficient instruc-
679 tions to faithfully reproduce the main experimental results, as described in supplemental
680 material?

681 Answer: [Yes]

682 Justification: Anonymized code/logs link in footnote 1; repository README documents
683 bootstrap, scripts, and `rebuild/NUMBERS.csv` as the frozen number authority.

684 **6. Experimental setting/details**

685 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-
686 rameters, how they were chosen, type of optimizer) necessary to understand the results?

687 Answer: [Yes]

688 Justification: Zero-shot $T=0$ CoT, model roster, bank sizes, and variant definitions are in
689 Sections 3–3.1 and Appendix B.

690 **7. Experiment statistical significance**

691 Question: Does the paper report error bars suitably and correctly defined or other appropri-
692 ate information about the statistical significance of the experiments?

693 Answer: [Yes]

694 Justification: Wilson CIs, Fisher exact tests, Wilcoxon/bootstrap where applicable; meth-
695 ods noted in captions and Appendix A.

696 **8. Experiments compute resources**

697 Question: For each experiment, does the paper provide sufficient information on the com-
698 puter resources (type of compute workers, memory, time of execution) needed to reproduce
699 the experiments?

700 Answer: [Yes]

701 Justification: Approximately 20,000 API calls; mechanistic pilots are local open-weight
702 runs (Appendix F); remaining API gap fill is credit-limited (Limitations).

703 **9. Code of ethics**

704 Question: Does the research conducted in the paper conform, in every respect, with the
705 NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

706 Answer: [Yes]

707 Justification: Evaluation of public APIs/open models on synthetic reasoning items; no hu-
708 man subjects; anonymity preserved for review.

709 **10. Broader impacts**

710 Question: Does the paper discuss both potential positive societal impacts and negative
711 societal impacts of the work performed?

712 Answer: [Yes]

713 Justification: Positive impact is more honest capability reporting beyond accuracy; negative
714 risk is over-interpreting triangulation labels as causal mechanism claims—we mark labels
715 as calibration points (Section 4.5).

716 **11. Safeguards**

717 Question: Does the paper describe safeguards that have been put in place for responsible
718 release of data or models that have a high risk for misuse (e.g., pre-trained language models,
719 image generators, or scraped datasets)?

720 Answer: [N/A]

721 Justification: No new generative model or scraped personal data is released; artefacts are
722 evaluation logs and derived metrics.

723 **12. Licenses for existing assets**

724 Question: Are the creators or original owners of assets (e.g., code, data, models), used in
725 the paper, properly credited and are the license and terms of use explicitly mentioned and
726 properly respected?

727 Answer: [Yes]

728 Justification: PlanBench, GSM-Symbolic, Infini-gram, and model providers are cited;
729 clone sources listed in the repository README.

730 **13. New assets**

731 Question: Are new assets introduced in the paper well documented and is the documenta-
732 tion provided alongside the assets?

733 Answer: [Yes]

734 Justification: Question banks, variant generators, probe scripts, and frozen number file are
735 documented in the repository and Appendix.

736 **14. Crowdsourcing and research with human subjects**

737 Question: For crowdsourcing experiments and research with human subjects, does the pa-
738 per include the full text of instructions given to participants and screenshots, if applicable,
739 as well as details about compensation (if any)?

740 Answer: [N/A]

741 Justification: No crowdsourcing or human subjects.

742 **15. Institutional review board (IRB) approvals or equivalent for research with human**
743 **subjects**

744 Question: Does the paper describe potential risks incurred by study participants, whether
745 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
746 approvals (or an equivalent approval/review based on the requirements of your country or
747 institution) were obtained?

748 Answer: [N/A]

749 Justification: No human subjects research.

750 **16. Declaration of LLM usage**

751 Question: Does the paper describe the usage of LLMs if it is an important, original, or
752 non-standard component of the core methods in this research? Note that if the LLM is used
753 only for writing, editing, or formatting purposes and does *not* impact the core methodology,
754 scientific rigor, or originality of the research, declaration is not required.

755 Answer: [Yes]

756 Justification: LLMs are the evaluation subjects (API and open-weight models under the
757 three probes); they are not used as a methodological component beyond being the systems
758 under test.